Ordered Boundary Identities
and Gamma-Kernel Reconstruction

Boundary localization, polylogarithmic cancellation, generalized Stieltjes jets, and Gamma kernels

ProveIt Contributors

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Research status. This article supplies ordinary mathematical proofs, exact finite algebra, and separately labelled numerical diagnostics. It is a continuation of the current manuscript and all seven incoming polylogarithm packages inspected at the repository revision printed below. Historical priority is not asserted for the new formulations. No proof-assistant formalization is claimed.

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Abstract

An ordered assignment of independently shifted finite-difference weights contains information that its determinantal symmetrization erases. We recover that information explicitly. For two particles, two finite summations by parts reduce the ordered double sum to a single two-shift zeta kernel. This gives its joint meromorphic continuation, every polar coefficient, all convergent positive-integer-order evaluations and their finite parts, a divided-difference digamma generating function, and a Gamma primitive. Oblique poles explain direction-dependent values at parameters where every pointwise weight vanishes. A cutoff generating function is then proved on an entire two-order parameter domain: its sole depth-two polylogarithm has the factor $(1-z)^2$, which explains the endpoint reduction. Beta parametrization yields ordinary convergent identities involving every generalized Stieltjes constant. At arbitrary particle number, transferring the stencils to the ordered Vandermonde square localizes the sum to collision blocks; an explicit finite algorithm uses at most $\lfloor (r+1)/2 \rfloor$ free summation indices. A separate Gamma-integral basis solves the fixed-gap null equation for unequal even harmonic tails. Its exact normalization and nonzero determinant asymptotic prove eventual optimal reconstruction on every fixed-gap ray; the first relative correction is an explicit harmonic-number expression. All formulas retain their convergence domains, confluence conventions, and finite-part coordinates. Exact algebraic tests and numerical checks accompany the proofs.

1 The source questions and the mathematical setting

1.1 A continuation of the current material

The source snapshot is the ProveIt revision `e3d7ec5d3f9be259c1f91edaea4389f4e79b5b03`, obtained on 11 October 2026. Both requested directories were inspected: the canonical manuscript and the seven then-present incoming archives. The manuscript's claim ledger remains the authority for accepted canonical results [1, 2]; an incoming report's delivery status does not by itself amount to canonical proof acceptance.

The most direct target is the incoming *Finite-Difference Determinants* report [4]. Its section entitled "Unsymmetrized mixed-order sums" asks which additional coordinates are needed for an individual ordering of independent spectral labels. Its existing polarization theorem gives only the sum over assignments. It also asks for a finite description of noninteger-order cutoff generators and for extensions to other annihilating stencils. These questions are mathematically related: shifting the finite differences onto the ordered summation domain exposes the boundaries at which the extra coordinates live.

The present two-particle result answers the individual-ordering question on an open complex domain, not only at isolated integer values. It shows that a one-index kernel is enough. That kernel has explicit Hurwitz expansions and, at positive integer orders, a finite polygamma evaluation. The all-particle result is different: it gives a finite localization formula and a bound on the number of free indices, rather than incorrectly extrapolating a universal single-zeta formula.

A second target comes from the harmonic-tail Gamma continuation [5]. Its exact fixed-gap reduction leaves an m-dimensional null equation, where the two tail orders differ by $2m$. Section 8 constructs a basis by ordinary integrals against $|\Gamma(\frac{1}{4}+\frac{ix}{2})|^{4m}$. The distinct decay scales of these integrals produce a nonzero limiting Vandermonde determinant. This proves that, for every fixed m, the first $2b$ moments form the optimal formal prefix for all sufficiently large $(p,q)=(2b-2m,2b)$. The theorem is complementary to the existing eventual result with the smaller order fixed [8]. Neither assertion alone settles every unequal pair.

The current sources also make clear why old open labels cannot simply be reused. Several finite-order harmonic-tail reconstruction cases have already received proofs in recent reports. The short Gaussian S_6 and revised S_8 evaluations, by contrast, remain conjectural in the

current canonical manuscript [2, 3]. None of the ordered identities below is represented as a proof of either Gaussian evaluation.

1.2 Classical ingredients and the scope of originality

Finite summation by parts, finite differences, partial fractions, Hurwitz continuation, and Gamma differentiation are classical. The general rational-summability role of discrete residues is treated by Chen–Singer and Arreche–Sitaula [14, 15]. Multivariable zeta functions can have intersecting polar hyperplanes and direction-sensitive Laurent behavior; this phenomenon is not new in itself [16]. Beta parametrization is also an established method for zeta products [17].

The contributions claimed here are the explicit ordered boundary identity, its complete consequences in this family, the finite cutoff polylogarithm formula and its endpoint cancellation, and the all-particle collision-block reduction. Their derivations were checked against the inspected sources. The literature search does not establish worldwide historical priority. In particular, finite polygamma reduction of a rational one-index summand is a classical step; it is the preceding reduction of the ordered ensemble that supplies the new application.

The Gamma basis, its normalization, and its fixed-gap consequences form the second group of contributions. Classical Fourier transforms of Gamma products provide useful context [18]; the result here concerns a specific higher-power weight, its sinh/tanh modes, and the repository's exact reconstruction determinant.

1.3 Notation and analytic conventions

For $\operatorname{Re} a > 0$ we use the principal logarithm in $(n+a)^{-s} = \exp(-s\log(n+a))$. Thus every shifted denominator in the original series stays in the right half-plane. Write

$$(s)_j = s(s+1)\cdots(s+j-1), \qquad (s)_0 = 1, \qquad \psi(a) = \frac{\Gamma'(a)}{\Gamma(a)}.$$

For positive integers p,

$$H_n^{(p)} = \sum_{k=1}^{n} k^{-p}, \qquad H_0^{(p)} = 0.$$

The same finite sum defines $H_n^{(s)}$ for complex s. We use the Stieltjes convention

$$\zeta(1+u,a) = \frac{1}{u} + \sum_{m=0}^{\infty} \frac{(-1)^m \gamma_m(a)}{m!} u^m, \qquad \gamma_0(a) = -\psi(a). \tag{1.1}$$

Derivatives of ζ without an argument qualifier differentiate its first variable. Standard relations used below include

$$\partial_a \zeta(s,a) = -s\zeta(s+1,a), \qquad \zeta(p,a) = \frac{(-1)^p}{(p-1)!}\psi^{(p-1)}(a) \quad (p \geq 2), \tag{1.2}$$

and $\zeta'(0,a) = \log\Gamma(a) - \frac{1}{2}\log(2\pi)$ [9, 11].

An equality continued from a normally convergent domain is always an equality of those continuations. It need not equal the result of substituting exceptional parameters into every summand first. Likewise, a finite part is the constant in a declared Laurent coordinate; a hard cutoff is a different operation unless their agreement is proved.

For $|z| < 1$ our multiple-polylogarithm convention is

$$\operatorname{Li}_{t,s}(z,1) = \sum_{k>j\geq 1} \frac{z^k}{k^t j^s}. \tag{1.3}$$

The exponential factor makes this entire in the two order variables. This is independent of the convergence restrictions of its value at $z = 1$.

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2 The ordered two-particle boundary identity

Define

$$w_n(s,a) = \frac{1}{2}\big((n+a)^{-s} - 2(n+a+1)^{-s} + (n+a+2)^{-s}\big) \tag{2.1}$$

and retain the ordering of its labels:

$$\mathcal{O}(s,t;a,b) = \sum_{0\le m<n} (n-m)^2 w_m(s,a) w_n(t,b). \tag{2.2}$$

In particular, exchanging (s,a) and (t,b) generally changes the answer. Let

$$\mathcal{K}(s,t;x,y) = \sum_{k=0}^{\infty} (k+x)^{-s}(k+y)^{-t}, \qquad \operatorname{Re}(s+t) > 1, \tag{2.3}$$

and set

$$E_s(a) = \frac{a^{-s} - (a+1)^{-s}}{2}. \tag{2.4}$$

Theorem 2.1 (Ordered boundary reduction)**.** *For*

$$\operatorname{Re} a, \operatorname{Re} b > 0, \qquad \operatorname{Re} t > 1, \qquad \operatorname{Re}(s+t) > 0, \tag{2.5}$$

the sum (2.2) *converges absolutely and locally normally, and*

$$\boxed{\mathcal{O}(s,t;a,b) = E_s(a)\zeta(t,b+1) + \frac{1}{4}\{\mathcal{K}(s,t;a+2,b+1) - \mathcal{K}(s,t;a,b+1)\}.} \tag{2.6}$$

On the portion $0 < \operatorname{Re}(s+t) \le 1$*, the kernel difference means the single combined convergent series*

$$\mathcal{D}(s,t;a,b) = \sum_{k=0}^{\infty} (k+b+1)^{-t}\{(k+a+2)^{-s} - (k+a)^{-s}\}. \tag{2.7}$$

All fixed finite derivatives in the four parameters can be taken termwise on compact subsets of (2.5).

2.1 Convergence on the asymmetric domain

Repeated integration of a second derivative gives

$$w_n(s,a) = \frac{s(s+1)}{2} \int_0^1 \int_0^1 (n+a+u+v)^{-s-2}\, \mathrm{d}u\, \mathrm{d}v. \tag{2.8}$$

Consequently $w_n(s,a) = O((n+1)^{-\operatorname{Re} s-2})$, uniformly on compact parameter sets, including neighborhoods of $s = 0, -1$. For a fixed outer index n, bound $(n-m)^2$ by n^2. The inner sum of $(m+1)^{-\operatorname{Re} s-2}$ is bounded when $\operatorname{Re} s > -1$, grows logarithmically at $\operatorname{Re} s = -1$, and is $O(n^{-\operatorname{Re} s-1})$ when $\operatorname{Re} s < -1$. Combining these bounds with the outer weight gives convergence under (2.5). On compact sets one can decrease the real exponents slightly to obtain a single summable majorant, so the transition at $\operatorname{Re} s = -1$ causes no loss of normality. Cauchy estimates on slightly larger compact neighborhoods justify every fixed derivative.

The first difference in (2.7) is $O((k+1)^{-\operatorname{Re} s-1})$. Thus its combined series has precisely the sufficient domain $\operatorname{Re}(s+t) > 0$ used in the theorem. No separate summation of divergent kernel terms is involved.

2.2 A finite identity with both upper endpoints

The proof works for arbitrary finite sequences before any special function is introduced. Put $f_n = (n+a)^{-s}$ and $g_n = (n+b)^{-t}$, and define $A_{-1} = A_0 = 0$ and

$$A_n = \sum_{m=0}^{n-1} (n-m)^2 (f_m - 2f_{m+1} + f_{m+2}). \tag{2.9}$$

Collecting the coefficient of each f_k gives, for $n \geq 1$,

$$A_n = n^2 f_0 + (-n^2 - 2n + 1) f_1 + 2 \sum_{k=2}^{n} f_k + f_{n+1}. \tag{2.10}$$

Taking two backward differences yields

$$A_n - 2A_{n-1} + A_{n-2} = 2(f_0 - f_1) - f_{n-1} + f_{n+1} \qquad (n \geq 1). \tag{2.11}$$

At $n = 1$ this is a direct check, using $A_{-1} = 0$.

Let $\mathcal{O}_N$ denote the restriction $n \leq N$ in (2.2). Shifting the three finite sums in $4\mathcal{O}_N = \sum_{n=1}^{N} A_n(g_n - 2g_{n+1} + g_{n+2})$ gives

$$\begin{aligned} 4\mathcal{O}_N = &\sum_{n=1}^{N} g_n \{2(f_0 - f_1) - f_{n-1} + f_{n+1}\} \\ &+ g_{N+1}(A_{N-1} - 2A_N) + g_{N+2} A_N. \end{aligned} \tag{2.12}$$

This is an exact finite identity. In particular, no endpoint term has been inferred from an asymptotic prescription.

Proof of theorem 2.1. The upper boundary in (2.12) can be grouped as

$$A_N(g_{N+2} - g_{N+1}) + g_{N+1}(A_{N-1} - A_N).$$

Formula (2.10) shows that A_N is bounded by a sum of $O(N^2)$ and $O(N^{1-\operatorname{Re} s})$, with a logarithmic enlargement at a borderline exponent. Its first backward difference is bounded by the corresponding $O(N)$ and $O(N^{-\operatorname{Re} s})$ terms. Since $g_{N+2} - g_{N+1} = O(N^{-\operatorname{Re} t - 1})$, the grouped boundary tends to zero under (2.5). The estimates are locally uniform with an arbitrarily small exponent margin. The remaining sum in (2.12) is $4E_s(a)\zeta(t, b+1) + \mathcal{D}(s,t;a,b)$. Division by four proves the theorem. □

2.3 What symmetrization removes

The existing polarized determinant evaluates $\frac{1}{2}(\mathcal{O}(s,t;a,b) + \mathcal{O}(t,s;b,a))$. The two one-index kernel differences satisfy the finite shift identity

$$\mathcal{D}(s,t;a,b) + \mathcal{D}(t,s;b,a) = -a^{-s}(b+1)^{-t} - (a+1)^{-s} b^{-t}. \tag{2.13}$$

Indeed, shift the first positive kernel once, pair it with the opposite negative kernel, and retain its missing zeroth term; do the same with the other pair. This is first legitimate in an absolute-convergence domain and then holds by continuation. Combining (2.13) with (2.6) recovers the existing polarized formula. It also identifies exactly which information the averaging has removed.

At a common shift and common order,

$$\mathcal{O}(s,s;a,a) = E_s(a)\zeta(s,a) - \frac{a^{-2s}}{2} + \frac{[a(a+1)]^{-s}}{4}. \tag{2.14}$$

This is the known two-particle specialization of the incoming single-zeta determinant [4]. Thus the present theorem extends that formula while preserving its normalization.

3 Joint continuation, oblique poles, and directional values

3.1 A convergent expansion in single Hurwitz zeta functions

Set $\delta = a - b$ and $y = b + 1$. For a sufficiently large integer M, expand each shifted factor about $k + y$ on the tail $k \geq M$. This gives

$$\mathcal{D}(s,t;a,b) = \sum_{k=0}^{M-1} (k+y)^{-t}\{(k+a+2)^{-s} - (k+a)^{-s}\} + \sum_{j=1}^{\infty} \frac{(-1)^j (s)_j}{j!}\{(\delta+1)^j - (\delta-1)^j\}\zeta(s+t+j, M+y). \tag{3.1}$$

Choose M so that both $|\delta + 1|$ and $|\delta - 1|$ are strictly smaller than $|M + y|$; on any compact set a uniform strict inequality is available. The factor $(s)_j/j!$ grows at most polynomially in j on compact s-sets, whereas the large-order Hurwitz series supplies geometric decay after multiplication by the displayed shift powers. The finitely many possible small-order zeta poles are handled separately. Hence the tail is locally normally convergent away from those poles, together with every fixed derivative. This proves joint meromorphic continuation to all $(s,t) \in \mathbb{C}^2$.

Theorem 3.1 (Complete polar data)**.** *The possible polar divisors of $\mathcal{O}$ are*

$$t = 1, \qquad s + t = 1 - j \quad (j = 1, 2, \ldots). \tag{3.2}$$

Away from intersections, their residues in the indicated linear coordinates are

$$\begin{aligned} \operatorname{Res}_{t=1} \mathcal{O} &= E_s(a), \\ \operatorname{Res}_{s+t=1-j} \mathcal{O} &= C_j(s;\delta) := \frac{(-1)^j (s)_j}{4j!}\{(\delta+1)^j - (\delta-1)^j\}. \end{aligned} \tag{3.3}$$

These coefficient formulas continue across intersections. Every local expression is a sum of terms with single linear denominators and a holomorphic remainder; no product denominator is required. A listed divisor can disappear on a special shift or order locus. The divisor $s+t=1$ is absent identically.

Proof. In (3.1) a pole occurs only when $s + t + j = 1$, with residue equal to its explicit coefficient, since Hurwitz zeta has residue one. The $j = 0$ binomial coefficient is zero in the difference, so its would-be divisor cancels before continuation. The other term in (2.6) has only $t = 1$ as a possible pole. Local normal convergence of the remaining tail proves the stated local decomposition. □

Corollary 3.2 (Cancellation in the averaged sum)**.** *Every oblique polar divisor cancels between $\mathcal{O}(s,t;a,b)$ and $\mathcal{O}(t,s;b,a)$.*

Proof. Equation (2.13) is entire in both orders. One can also check residues directly: on $s + t = 1 - j$, $(t)_j = (-1)^j (s)_j$, and replacing δ by $-\delta$ reverses the required coefficient. Thus the cancellation is an exact functional identity, not a numerical relation between periods. □

3.2 Four intersections at which every weight vanishes

Both $w_n(0,a)$ and $w_n(-1,a)$ vanish for every n. The continued ordered sum nevertheless has directional values at pairs of these parameters. The origin gives the simplest explanation.

Theorem 3.3 (Universal origin behavior)**.** *Near* $(s,t)=(0,0)$*,*

$$\mathcal{O}(s,t;a,b)=-\frac{s}{2(s+t)}+H(s,t;a,b),\qquad H\ \textit{holomorphic},\qquad H(0,0;a,b)=0. \tag{3.4}$$

Consequently, if $\alpha+\beta\neq 0$*,*

$$\lim_{\varepsilon\to 0}\mathcal{O}(\alpha\varepsilon,\beta\varepsilon;a,b)=-\frac{\alpha}{2(\alpha+\beta)}. \tag{3.5}$$

The limit is independent of both spatial shifts.

Proof. The $j=1$ coefficient in (3.3) is exactly $-s/2$. All other divisors miss a sufficiently small neighborhood of the origin. After subtracting this polar term, the remainder is holomorphic. On the punctured slice $s=0$ the explicit formula (2.6) vanishes identically, so the holomorphic remainder vanishes at the origin. Substitution of a non-tangent line proves (3.5). □

The diagonal value $-1/4$ recovers the specialization anomaly already recorded in [4]. Formula (3.5) is more informative: the slice $s=0$ has limit zero, the slice $t=0$ has limit $-1/2$, and general slopes interpolate rationally. These are limits of the same meromorphic continuation, not different summation rules chosen for the pointwise zero series.

Theorem 3.4 (The complete 2×2 zero-weight table)**.** *Let* $m,n\in\{0,1\}$ *and* $\alpha+\beta\neq 0$*. Along* $s=-m+\alpha\varepsilon$*,* $t=-n+\beta\varepsilon$*,*

$$\lim_{\varepsilon\to 0}\mathcal{O}(s,t;a,b)=\frac{\alpha}{\alpha+\beta}\,c_{m,n}(\delta), \tag{3.6}$$

where

$c_{m,n}(\delta)$	$n=0$	$n=1$
$m=0$	$-\frac{1}{2}$	$\frac{\delta}{2}$
$m=1$	$-\frac{\delta}{2}$	$\frac{3\delta^2+1}{12}$

. (3.7)

Proof. Only $j=m+n+1$ supplies a divisor through the given point. Its coefficient $C_j(s;\delta)$ vanishes at $s=-m$ and has derivative $c_{m,n}(\delta)$ there. The first three polynomials are

$$C_1=-\frac{s}{2},\qquad C_2=\frac{\delta s(s+1)}{2},\qquad C_3=-\frac{(3\delta^2+1)s(s+1)(s+2)}{12}.$$

The restrictions $\mathcal{O}(0,t;a,b)$ and $\mathcal{O}(-1,t;a,b)$ are identically zero meromorphic functions of t, directly from (2.6). Thus the holomorphic remainder at the chosen point vanishes. Expanding the one polar quotient proves the table and the directional limit. □

No limit is asserted for directions with $\alpha+\beta=0$. They lie on the polar hyperplane itself, where cancellation of a numerator at one point does not define a restriction of the meromorphic function to the whole line.

4 Finite polygamma evaluation and a digamma master

The one-index reduction is useful only if its new kernel can be handled explicitly. At integer orders it is finite; at confluence its apparent singularities disappear; and the entire positive-order family is generated by one function of two shifts.

4.1 Partial fractions with all simple poles paired

Proposition 4.1. *Let p, q be positive integers and $\operatorname{Re} x, \operatorname{Re} y > 0$. If $x \neq y$, put $D = y - x$ and*

$$A_k = (-1)^{p-k}\binom{p+q-k-1}{q-1}D^{-(p+q-k)}, \quad 1 \le k \le p, \tag{4.1}$$

$$B_l = (-1)^{p}\binom{p+q-l-1}{p-1}D^{-(p+q-l)}, \quad 1 \le l \le q. \tag{4.2}$$

Then $A_1 + B_1 = 0$ and

$$\mathcal{K}(p, q; x, y) = A_1\{\psi(y) - \psi(x)\} + \sum_{k=2}^{p} A_k \zeta(k, x) + \sum_{l=2}^{q} B_l \zeta(l, y). \tag{4.3}$$

The empty sums are zero. At $x = y$ the value is $\zeta(p + q, x)$.

Proof. Expand $(X + D)^{-q}$ at $X = 0$, and X^{-p} at $X = -D$. The respective principal parts of $X^{-p}(X + D)^{-q}$ have coefficients (4.1) and (4.2). Their difference from that rational function is entire and vanishes at infinity, hence is zero. Its X^{-1} coefficient at infinity gives $A_1 + B_1 = 0$. Consequently the only two series which would diverge separately occur as the convergent pair

$$A_1 \sum_{n \ge 0}\left(\frac{1}{n + x} - \frac{1}{n + y}\right) = A_1\{\psi(y) - \psi(x)\}.$$

All higher powers sum absolutely. This proves (4.3). The confluent value follows directly from the original kernel. □

More generally, confluence comes with every Taylor coefficient:

$$\mathcal{K}(s, t; x, x + h) = \sum_{r=0}^{\infty} \frac{(-1)^r (t)_r}{r!} h^r \zeta(s + t + r, x), \qquad |h| < |x|. \tag{4.4}$$

Here $\operatorname{Re} x > 0$ and $\operatorname{Re}(x + h) > 0$; on compact sets a uniformly smaller disk is understood. The identity follows first for $\operatorname{Re}(s + t) > 1$ by the binomial theorem, since $|n + x| \ge |x|$, and then meromorphically. It is preferable to subtracting nearly cancelling negative powers of D when computing close shifts. The finite formula and (1.2) establish a finite polygamma evaluation of every $\mathcal{O}(p, q; a, b)$ with $p \ge 1$, $q \ge 2$.

4.2 The exact parity left by equal shifts

Theorem 4.2. *For $p \ge 1$, $q \ge 2$, $N = p + q$ and $\operatorname{Re} a > 0$, define*

$$L_j = \begin{cases} \binom{N-j-1}{q-1}, & j \le p, \\ 0, & j > p, \end{cases} \qquad R_j = \begin{cases} \binom{N-j-1}{p-1}, & j \le q, \\ 0, & j > q. \end{cases}$$

Then

$$\begin{aligned} \mathcal{O}(p, q; a, a) = {} & E_p(a)\zeta(q, a + 1) \\ & + \frac{(-1)^q}{4} \sum_{j=2}^{\max(p,q)} \{1 - (-1)^{N-j}\}\{L_j + (-1)^j R_j\}\zeta(j, a + 1) \\ & - \frac{1}{4}\sum_{k=1}^{p}\binom{N-k-1}{q-1}\{(-1)^q (a + 1)^{-k} + (-1)^{p-k} a^{-k}\}. \end{aligned} \tag{4.5}$$

Only orders j with $p + q + j$ odd occur in the correction in the second line. When $p = q$, that entire line vanishes.

Proof. Apply (4.3) to $\mathcal{K}(p,q;a+2,a+1)-\mathcal{K}(p,q;a,a+1)$, so that $D=-1$ and $D=1$. Use $\zeta(j,a)=a^{-j}+\zeta(j,a+1)$ and $\zeta(j,a+2)=\zeta(j,a+1)-(a+1)^{-j}$, together with the same shift recurrence for ψ. For $j\geq 2$ the resulting coefficient is $(-1)^q\{1-(-1)^{N-j}\}\{L_j+(-1)^jR_j\}/4$. The simple-pole pairs contribute only the finite third line. If $p=q$, then $L_j=R_j$: the second brace vanishes for odd j, and the first vanishes for even j. □

This is a statement about explicit coefficients. It makes no claim that the special values remaining in a particular formula are arithmetically independent.

Example 4.3 (The ordering survives)**.** Two equal-shift evaluations are

$$\mathcal{O}(2,3;1,1)=\frac{3}{8}\zeta(3)-\frac{3}{2}\zeta(2)+\frac{33}{16}, \tag{4.6}$$

$$\mathcal{O}(3,2;1,1)=\frac{31}{16}\zeta(2)-\frac{95}{32}. \tag{4.7}$$

Their average is the previously polarized example $7\zeta(2)/32+3\zeta(3)/16-29/64$ in [4]. Confluent shifts give a different cancellation:

$$\mathcal{O}(2,2;1,2)=\frac{1}{4}\zeta(4)+\frac{1}{4}\zeta(2)-\frac{9}{16}, \tag{4.8}$$

$$\mathcal{O}(2,2;2,1)=\frac{7}{36}\zeta(2)-\frac{1}{4}\zeta(4)-\frac{11}{576}. \tag{4.9}$$

The $\zeta(4)$ term is an exact confluence contribution erased when the two assignments are added.

4.3 One master function generates all positive orders

Define the holomorphic divided difference

$$\Psi[x,y]=\begin{cases}\dfrac{\psi(y)-\psi(x)}{y-x}, & x\neq y,\\ \psi'(x), & x=y.\end{cases}$$

For $\operatorname{Re}a,\operatorname{Re}b>0$ set

$$F(a,b)=-\frac{\psi(b+1)}{2a(a+1)}+\frac{1}{4}\{\Psi[a+2,b+1]-\Psi[a,b+1]\}. \tag{4.10}$$

Theorem 4.4 (Digamma master)**.** *For $p\geq 1$ and $q\geq 2$,*

$$\mathcal{O}(p,q;a,b)=\frac{(-1)^{p+q-2}}{(p-1)!(q-1)!}\partial_a^{p-1}\partial_b^{q-1}F(a,b). \tag{4.11}$$

For real $a,b>0$ and $|x|<a$, $|y|<b$ this is equivalently

$$\sum_{p\geq 1}\sum_{q\geq 2}x^{p-1}y^{q-1}\mathcal{O}(p,q;a,b)=F(a-x,b-y)-F(a-x,b). \tag{4.12}$$

Both apparent singularities $b=a\pm 1$ in a quotient version of F are removable.

Proof. The residue of $\mathcal{O}(1,t;a,b)$ at $t=1$ is $E_1(a)=1/(2a(a+1))$. The constant of the Hurwitz zeta factor there is $-\psi(b+1)$, while $\mathcal{K}(1,1;x,y)=\Psi[x,y]$ is convergent. Thus (4.10) is precisely $\mathrm{FP}_{t=1}\,\mathcal{O}(1,t;a,b)$ with a,b fixed.

Direct differentiation of the weights gives

$$\partial_a^k w_n(s,a)=(-1)^k(s)_k w_n(s+k,a). \tag{4.13}$$

For $q \geq 2$,

$$\partial_b^{q-1}\mathcal{O}(1,t;a,b) = (-1)^{q-1}(t)_{q-1}\mathcal{O}(1,t+q-1;a,b).$$

The b derivative kills the residue, and the right side is regular at $t=1$. Its value is $(-1)^{q-1}(q-1)!\mathcal{O}(1,q;a,b)$. Differentiate in a and use (4.13) again. This proves (4.11); Taylor expansion proves (4.12).

There is also a direct convergence interpretation. Summing the geometric weight expansions produces $w_m(1,a-x)$ in the inner index and $w_n(1,b-y)-w_n(1,b)$ in the outer index. The latter difference has one additional power of decay. Their original ordered sum is absolutely convergent on the stated polydisk. This proves that (4.12) is an ordinary generating identity. □

The slightly larger domain $a > 0$, $b > -1$, $|x| < a$, $|y| < b+1$ also works: the outer summation starts at $n=1$. We retain the smaller common shift domain elsewhere to keep the original definitions uniform.

4.4 Nonlinear harmonic weights and Gamma ratios

Let $\mathcal{B}_r$ denote the complete exponential Bell polynomial, normalized by

$$\exp\left(\sum_{j\geq 1} x_j \frac{v^j}{j!}\right) = \sum_{r\geq 0} \mathcal{B}_r(x_1,\dots,x_r)\frac{v^r}{r!}. \tag{4.14}$$

For $h_j(n,a) = \sum_{\ell=0}^{2}(n+a+\ell)^{-j}$, one has

$$w_n(p,a) = \frac{\Gamma(n+a)}{\Gamma(n+a+3)}\frac{\mathcal{B}_{p-1}(h_1, 1!h_2, \dots, (p-2)!h_{p-1})}{(p-1)!}. \tag{4.15}$$

For $p=1$ the Bell polynomial is $\mathcal{B}_0 = 1$. Indeed $w_n(1,a) = \Gamma(n+a)/\Gamma(n+a+3)$, and the logarithmic derivatives of this ratio are $(-1)^j(j-1)!h_j$. Equation (4.13) and the Bell formula for successive derivatives prove (4.15).

This translates the ordered identities into explicit nonlinear harmonic sums. For example, write $P_n = (n+1)(n+2)(n+3)$ and $h_{j,n} = H_{n+3}^{(j)} - H_n^{(j)}$. Then (4.6) becomes

$$\sum_{0\leq m<n} \frac{(n-m)^2 h_{1,m}(h_{1,n}^2 + h_{2,n})}{P_m P_n} = \frac{3}{4}\zeta(3) - 3\zeta(2) + \frac{33}{8}. \tag{4.16}$$

All higher Bell combinations are supplied by the same finite master, with no new summation coordinates. At integral shifts they involve finite harmonic differences; at arbitrary shifts the h_j are finite polygamma differences by the usual recurrence.

4.5 A Gamma primitive under common translation

For $a, b > 0$, $h \geq 0$, define $\Lambda_h(x) = \log\Gamma(x+h) - \log\Gamma(x)$. Then

$$\begin{aligned}\int_0^h \mathcal{D}(1,1;a+c,b+c)\,\mathrm{d}c = &\frac{\Lambda_h(b+1)-\Lambda_h(a+2)}{b-a-1}\\ &-\frac{\Lambda_h(b+1)-\Lambda_h(a)}{b-a+1}.\end{aligned} \tag{4.17}$$

At a zero denominator its quotient is the confluent derivative $\Lambda_h'(x) = \psi(x+h)-\psi(x)$. To prove the identity, replace each $\mathcal{K}(1,1)$ by its digamma divided difference; common translation leaves its denominator constant, and the primitive of ψ is $\log\Gamma$. This also proves the holomorphic path version in the right half-plane. For positive integral h, the Gamma ratios are finite products, so the primitive becomes a finite logarithmic identity. Further argument derivatives generate polygamma primitives of all positive-order kernels.

5 The cutoff polylogarithm and its disappearing depth

The ordered endpoint evaluation is accompanied by an explicit cutoff identity. This is useful both as a proof of the endpoint cancellation and as a finite transform for noninteger orders, one of the questions raised in [4].

5.1 An identity entire in the two orders

For $|z| < 1$ put

$$G_{s,t}(z) = \sum_{0 \le m < n} (n-m)^2 w_m(s,1) w_n(t,1) z^{n+1}, \tag{5.1}$$

$$B_{s,t}(z) = \sum_{k \ge 1} \frac{z^k}{k^s (k+1)^t}, \qquad c_s = \frac{1-2^{-s}}{2}, \qquad \Theta = z\frac{\mathrm{d}}{\mathrm{d}z}, \tag{5.2}$$

$$U_t(z) = \frac{(1-z)^2}{z^2} \operatorname{Li}_t(z) + 2 - z^{-1} - 2^{-t}. \tag{5.3}$$

The last expression has a removable singularity at zero and equals $2\sum_{n\ge 0} w_n(t,1) z^{n+1}$. The same convention will apply to the removable negative powers of z in the next formula.

Theorem 5.1 (Finite cutoff formula)**.** *For all $s, t \in \mathbb{C}$ and $|z| < 1$,*

$$\begin{aligned} G_{s,t}(z) = {} & \frac{(1-z)^2}{2z^2} \operatorname{Li}_{t,s}(z,1) + \frac{z-1}{2z} \operatorname{Li}_{s+t}(z) + \frac{1}{4} B_{t,s}(z) - \frac{1}{4z} B_{s,t}(z) \\ & + \frac{1}{2}\left(c_s \Theta^2 - \Theta - \frac{1}{2}\right) U_t(z) + \frac{1}{2} - 2^{-t-2}. \end{aligned} \tag{5.4}$$

Every term is entire in s, t on the open unit disk. At positive integer orders the two B functions are finite combinations of ordinary polylogarithms. The only depth-two term in this representation has the explicit factor $(1-z)^2$.

Proof. The inner finite identity of Section 2, with $a = 1$, becomes

$$\sum_{m=0}^{n-1} (n-m)^2 w_m(s,1) = H_{n+1}^{(s)} + c_s (n+1)^2 - (n+1) - \frac{1}{2} + \frac{1}{2(n+2)^s}. \tag{5.5}$$

It also holds at $n = 0$, since the right side then vanishes. For $k \ge 1$, the three generating identities needed for its harmonic part are

$$\begin{aligned} \sum_{k\ge 1} \frac{z^k H_k^{(s)}}{k^t} &= \operatorname{Li}_{t,s}(z,1) + \operatorname{Li}_{s+t}(z), \\ \sum_{k\ge 1} \frac{z^k H_k^{(s)}}{(k+1)^t} &= z^{-1} \operatorname{Li}_{t,s}(z,1), \\ \sum_{k\ge 1} \frac{z^k H_k^{(s)}}{(k+2)^t} &= z^{-2} \operatorname{Li}_{t,s}(z,1) - z^{-1} B_{s,t}(z). \end{aligned} \tag{5.6}$$

They follow by writing the finite harmonic sum and changing the outer index; the term subtracted in the last line is precisely the last inner index in the shifted nested sum.

Multiply (5.5) by $w_n(t,1) z^{n+1}$ and sum. Its quadratic polynomial contributes $\frac{1}{2}(c_s\Theta^2 - \Theta - \frac{1}{2})U_t$. The harmonic contribution, by (5.6), is

$$\frac{(1-z)^2}{2z^2} \operatorname{Li}_{t,s}(z,1) + \frac{1}{2} \operatorname{Li}_{s+t}(z) - \frac{1}{2z} B_{s,t}(z).$$

The final term $1/(2(n+2)^s)$ contributes

$$\frac{1}{4}B_{t,s}(z) - \frac{1}{2z}\operatorname{Li}_{s+t}(z) + \frac{1}{4z}B_{s,t}(z) + \frac{1}{2} - 2^{-t-2}.$$

Adding gives (5.4). On compact sets of orders, every summand grows at most polynomially in its indices before the exponential factor is applied. All rearrangements therefore converge normally and prove the identity for every pair of complex orders. □

5.2 Finite classical polylogarithms at integer orders

For positive integers p, q, use the coefficients of (4.1)–(4.2) with $D = 1$. Then

$$B_{p,q}(z) = \sum_{k=1}^{p} A_k \operatorname{Li}_k(z) + \sum_{l=1}^{q} B_l z^{-1}\{\operatorname{Li}_l(z) - z\}. \tag{5.7}$$

This follows by summing the rational partial fractions against z^n. At $|z| < 1$ the two simple-pole sums need not be paired for convergence, although pairing is necessary before taking their individual endpoint values. Since $\Theta \operatorname{Li}_\nu = \operatorname{Li}_{\nu-1}$, the differential term in (5.4) is also finite in classical polylogarithms. Thus the integer-order generator is explicitly in depth at most two, and its depth-two coefficient is known exactly.

Corollary 5.2 (Endpoint cancellation)**.** *For positive integers $p \geq 1$, $q \geq 2$, the radial limit is*

$$\mathcal{O}(p, q; 1, 1) = c_p\zeta(q) - \frac{1}{2} + 2^{-p-2} + \frac{1}{4}\{B_{q,p}(1) - B_{p,q}(1)\}. \tag{5.8}$$

Together with (5.7), interpreted through its paired simple poles, this yields the parity formula (4.5) at $a = 1$.

Proof. Here $\operatorname{Li}_{q,p}(1, 1)$ converges: the inner harmonic sum is bounded for $p > 1$ and grows only logarithmically for $p = 1$, whereas $q \geq 2$. Its factor $(1 - z)^2$ makes its contribution zero. The factor $z - 1$ similarly kills $\operatorname{Li}_{p+q}(z)$. Finite summation by parts for the three moments of the single weight gives

$$U_q(1) = 1 - 2^{-q}, \qquad (\Theta U_q)(1) = 1, \qquad (\Theta^2 U_q)(1) = 2\zeta(q) - 1.$$

These moments converge for $q > 1$. Substitution into (5.4) gives (5.8). Absolute convergence of the original double sum justifies its radial limit as well. □

There is a broader analytic interpretation. If $\operatorname{Re} t > 1$ and $\operatorname{Re}(s + t) > 0$, the original $G_{s,t}$ has the endpoint $\mathcal{O}(s, t; 1, 1)$. The depth-two contribution still tends to zero radially. Indeed $\sum_{j<k} j^{-\operatorname{Re} s}$ is $O(1 + k^{\max(1-\operatorname{Re} s,0)} \log(k + 1))$, and a direct comparison of the resulting power series with a Laplace integral proves $(1 - z)^2 \operatorname{Li}_{t,s}(z, 1) \to 0$ on this domain. Some of the remaining separated terms can diverge and must be kept in the combination in (5.4). The identity of the collective endpoint with the boundary formula is valid; replacing a divergent component by an unannounced regularized value is unnecessary.

5.3 A finite noninteger-order transform

Writing $B_{s,t}$ as a one-index series already reduces depth, but a standard special-function transform makes the answer more explicit. Let

$$\Phi(z, v, c) = \sum_{k\geq 0} \frac{z^k}{(k + c)^v}, \qquad |z| < 1,$$

be the Lerch transcendent [10]. For $\operatorname{Re} s, \operatorname{Re} t > 0$,

$$B_{s,t}(z) = \frac{z\Gamma(s+t)}{\Gamma(s)\Gamma(t)} \int_0^1 u^{s-1}(1-u)^{t-1}\Phi(z, s+t, 2-u)\,\mathrm{d}u. \tag{5.9}$$

To prove it, apply the beta identity

$$X^{-s}Y^{-t} = \frac{\Gamma(s+t)}{\Gamma(s)\Gamma(t)} \int_0^1 \frac{u^{s-1}(1-u)^{t-1}}{[uX+(1-u)Y]^{s+t}}\,\mathrm{d}u \tag{5.10}$$

with $X = k$, $Y = k+1$ and sum z^k. The initial exchange is absolutely convergent. Both sides then continue in their order variables from that domain; the continued B itself is entire for $|z| < 1$. Equations (5.4) and (5.9) give a finite description of the cutoff generator for noninteger orders using ordinary polylogarithms, one depth-two polylogarithm, and two ordinary beta–Lerch transforms. No assumption that arbitrary complex orders have finite integer partial fractions is made.

5.4 Euler primitives

For a function vanishing at zero, define $\mathcal{J}f(z) = \int_0^z f(u)\,\mathrm{d}u/u$ within the unit disk. Termwise integration yields, for every integer $\ell \geq 0$,

$$\mathcal{J}^\ell G_{s,t}(z) = \sum_{0\leq m<n} \frac{(n-m)^2}{(n+1)^\ell} w_m(s,1)w_n(t,1)z^{n+1}. \tag{5.11}$$

At positive integer orders these remain finite combinations of depth-at-most-two polylogarithms and rational functions. One constructive proof uses (5.5): the coefficient is a rational function of $k = n+1$, times $H_k^{(p)}$, plus a rational function. Dividing by k^ℓ introduces only additional integer-order poles at $0, -1, -2$. Partial fractions reduce each term to one of (5.6) with shifted orders, or to its finite index-shift analogues. Shifting $H_{k+c}^{(p)}$ changes it by a finite rational sum, so no further nested index is introduced. This proves closure under these primitives constructively; it does not assert a minimal transcendental basis at the endpoint.

6 Ordinary Stieltjes identities and their primitives

Two complementary constructions produce every generalized Stieltjes constant. The first subtracts a single asymptotic term from the original ordered summand and retains ordinary convergence. The second takes derivatives of a beta transform precisely on a hyperplane where the two Hurwitz residues cancel.

6.1 Subtracting before summing

For $\operatorname{Re} s > 0$ and $\operatorname{Re} a, \operatorname{Re} b > 0$, put

$$I_n(s,a) = \sum_{m=0}^{n-1} (n-m)^2 w_m(s,a), \qquad c_2(t) = \frac{t(t+1)}{2}.$$

Define the bracketed series

$$\mathcal{U}(s,t;a,b) = \sum_{n=1}^{\infty} \left[w_n(t,b)I_n(s,a) - E_s(a)c_2(t)(n+b)^{-t} \right]. \tag{6.1}$$

The subtraction is inside each summand.

Theorem 6.1 (Convergent Stieltjes jets)**.** *The series* (6.1) *is locally normally convergent for* $\operatorname{Re} s > 0$, $\operatorname{Re} t > 0$. *On this domain,*

$$\mathcal{U}(s,t;a,b) = \frac{1}{4}\mathcal{D}(s,t;a,b) + (1-c_2(t))E_s(a)\zeta(t,b+1), \tag{6.2}$$

with a removable singularity at $t=1$. *For* $m \geq 0$ *define the ordinarily convergent logarithmic difference*

$$J_m(s;a,b) = \sum_{k=0}^{\infty} \frac{\{(k+a+2)^{-s} - (k+a)^{-s}\}\log^m(k+b+1)}{k+b+1}. \tag{6.3}$$

With $\mathcal{U}_m = \partial_t^m \mathcal{U}(s,1;a,b)$ *and* $E = E_s(a)$, *the exact identities are*

$$\mathcal{U}_0 = \frac{1}{4}J_0 - \frac{3}{2}E, \tag{6.4}$$

$$\mathcal{U}_1 = -\frac{1}{4}J_1 - \frac{E}{2}\{1 + 3\gamma_0(b+1)\}, \tag{6.5}$$

$$\mathcal{U}_m = (-1)^m\left[\frac{1}{4}J_m + \frac{3m}{2}E\gamma_{m-1}(b+1) - \frac{m(m-1)}{2}E\gamma_{m-2}(b+1)\right], \quad m \geq 2. \tag{6.6}$$

Every $\mathcal{U}_m$ *on the left is obtained by differentiating the bracketed original series term by term.*

Proof. The finite identity for I_n gives $I_n(s,a) = E_s(a)(n+b)^2 + O(n)$ locally uniformly for $\operatorname{Re} s > 0$. For the second difference,

$$w_n(t,b) = c_2(t)(n+b)^{-t-2} + O(n^{-\operatorname{Re} t - 3})$$

uniformly on compact parameter sets. Their product after the subtraction is $O(n^{-\operatorname{Re} t - 1})$; any fixed order derivative only introduces powers of $\log n$, and a small compact-set margin absorbs them. Thus the series is locally normal.

For $\operatorname{Re} t > 1$, subtract the Hurwitz series from the ordered boundary identity. This proves (6.2) there and hence throughout the stated domain by analytic continuation. At $t = 1+v$,

$$1 - c_2(1+v) = -\frac{3}{2}v - \frac{1}{2}v^2.$$

Multiply this polynomial by (1.1). The mth derivative of $\mathcal{D}(s,t;a,b)$ at $t=1$ is $(-1)^m J_m$, justified by its normally convergent combined series. Extracting the constant, the linear coefficient, and the general coefficient proves (6.4)–(6.6). □

These identities provide a uniform way of expressing Stieltjes constants through logarithmically weighted ordered harmonic kernels. They are not divergent double sums assigned values after the fact: their defining brackets are summable before any continuation is used.

Example 6.2 (An explicit nonlinear harmonic subtraction)**.** At $a = b = 1$, $s = 2$, the values are

$$E = \frac{3}{8}, \qquad J_0 = \frac{11}{4} - 2\zeta(2), \qquad \mathcal{U}_0 = \frac{1}{8} - \frac{1}{2}\zeta(2).$$

For $s = 3$ one instead obtains $J_0 = 15/8 - 2\zeta(3)$ and $\mathcal{U}_0 = -3/16 - \zeta(3)/2$. In both cases the left side of (6.1) is explicit in the harmonic Bell weights (4.15).

6.2 The hard cutoff has a different constant

Let $\mathcal{O}_N$ denote the original ordered sum with $n \le N$. For $\operatorname{Re} s > 0$,

$$\lim_{N\to\infty}\{\mathcal{O}_N(s,1;a,b) - E_s(a)\log(N+b)\} = \operatorname{FP}_{t=1}\mathcal{O}(s,t;a,b) - \frac{3}{2}E_s(a), \tag{6.7}$$

where the finite part keeps s, a, b fixed. To prove it directly, write the finite sum as $\mathcal{U}_N(s,1;a,b)+E_s(a)\sum_{n=1}^{N}(n+b)^{-1}$. The first term tends to (6.4); the second, after subtracting $E_s(a)\log(N+b)$, tends to $-E_s(a)\psi(b+1)$. But

$$\operatorname{FP}_{t=1}\mathcal{O}(s,t;a,b) = -E_s(a)\psi(b+1) + \frac{1}{4}J_0(s;a,b). \tag{6.8}$$

This proves (6.7). Equivalently, the upper endpoints in the exact finite summation by parts have limit $-3E_s(a)/2$.

For example,

$$F(1,1) = \frac{\gamma}{4} - \frac{3}{8}, \qquad \lim_{N\to\infty}\left\{\mathcal{O}_N(1,1;1,1) - \frac{1}{4}\log(N+1)\right\} = \frac{\gamma}{4} - \frac{3}{4}.$$

There is no contradiction: the two operations retain different boundary constants. A diagonal Laurent coordinate $s = t$ would also differentiate the residue $E_s(a)$ and is yet another declared coordinate. These distinctions should be preserved in any integration of the formulas into the manuscript.

6.3 A beta transform on the cancelled polar hyperplane

For $\operatorname{Re} x, \operatorname{Re} y > 0$, $\operatorname{Re} s, \operatorname{Re} t > 0$ and initially $\operatorname{Re}(s+t) > 1$, summing (5.10) gives

$$\mathcal{K}(s,t;x,y) = \frac{\Gamma(s+t)}{\Gamma(s)\Gamma(t)}\int_0^1 u^{s-1}(1-u)^{t-1}\zeta(s+t, y+(x-y)u)\,\mathrm{d}u. \tag{6.9}$$

For $\mathcal{D}$ take the difference of the two integrands before continuing. The Hurwitz pole at $s+t=1$ then cancels identically. This is a direct instance of the beta-parametrization method for zeta products; compare [17]. The following all-order Stieltjes identity uses that cancellation.

Theorem 6.3 (Beta–Stieltjes difference identity)**.** *Suppose* $0 < \operatorname{Re}\sigma < 1$ *and* $\operatorname{Re} a, \operatorname{Re} b > 0$. *Set*

$$A_\pm(u) = b + 1 + (a - b \pm 1)u,$$
$$T_m(\sigma;a,b) = \sum_{k=0}^{\infty}\{(k+a+2)^{-\sigma} - (k+a)^{-\sigma}\}(k+b+1)^{\sigma-1}[-\log(k+b+1)]^m,$$

and define

$$\ell_1(u) = \log(1-u) + \psi(1) - \psi(1-\sigma),$$
$$\ell_j = \psi^{(j-1)}(1) - \psi^{(j-1)}(1-\sigma), \qquad j \ge 2. \tag{6.10}$$

Then, for every integer $m \ge 0$,

$$T_m(\sigma;a,b) = \frac{\sin(\pi\sigma)}{\pi}\int_0^1 u^{\sigma-1}(1-u)^{-\sigma} \times \sum_{j=0}^{m}\binom{m}{j}(-1)^j\mathcal{B}_{m-j}(\ell_1(u),\dots,\ell_{m-j})\{\gamma_j(A_+(u)) - \gamma_j(A_-(u))\}\,\mathrm{d}u. \tag{6.11}$$

Both the sum and the integral converge ordinarily. The sum has summands $O(k^{-2}\log^m k)$.

Proof. In the difference of (6.9), put $s = \sigma$, $t = 1 - \sigma + v$. Its integral is

$$\frac{\Gamma(1+v)}{\Gamma(\sigma)\Gamma(1-\sigma+v)} \int_0^1 u^{\sigma-1}(1-u)^{-\sigma+v}\{\zeta(1+v, A_+(u)) - \zeta(1+v, A_-(u))\}\,\mathrm{d}u.$$

The expression in braces is holomorphic at $v = 0$; its jth derivative there is $(-1)^j(\gamma_j(A_+) - \gamma_j(A_-))$. The factor outside the braces, divided by its value at $v = 0$, has logarithmic derivatives (6.10). The constant factor is $\sin(\pi\sigma)/\pi$ by Gamma reflection. The Bell product rule (4.14) therefore gives (6.11).

All endpoint powers have real exponents greater than -1, uniformly for v in a sufficiently small disk. Additional logarithms are integrable. The shifts $A_\pm(u)$ remain in a compact subset of the right half-plane, so the differentiated Hurwitz difference is bounded there. These facts justify differentiation under the integral. On the series side, the shifted difference gains one power of k, giving $O(k^{-2+|\mathrm{Re}\, v|}\log^m k)$ uniformly near zero. Differentiating it gives exactly T_m. □

At $\sigma = \frac{1}{2}$,

$$\ell_1(u) = \log[4(1-u)], \qquad \ell_j = (-1)^j(j-1)!(2-2^j)\zeta(j), \quad j \geq 2. \tag{6.12}$$

Thus the transform has an arcsine weight and Bell coefficients expressed entirely in ordinary zeta values. At $a = b = 1$ there is also the geometrically convergent single-zeta-derivative series

$$T_m(\sigma; 1, 1) = -2\sum_{h=0}^{\infty} \frac{(\sigma)_{2h+1}}{(2h+1)!}\zeta^{(m)}(2h+2, 2). \tag{6.13}$$

Indeed the Hurwitz expansion of $\mathcal{D}$ has only odd shift powers when $a = b$, and along $s+t = 1+v$ its zeta argument is $2h+2+v$. The shift-to-tail ratio is $1/2$, so differentiating this expansion is locally normal. In particular, $T_0(\frac{1}{2}; 1, 1) = -1/\sqrt{2}$; the next subsection explains its elementary telescoping origin.

6.4 Arcsine Gamma identities and Stieltjes primitives

Some symmetric members simplify further and should be identified as recurrence consequences, rather than presented as independent priority claims. For $a > 0$,

$$\int_0^1 \frac{\psi(a+1+u) - \psi(a+1-u)}{\sqrt{u(1-u)}}\,\mathrm{d}u = \frac{\pi}{\sqrt{a(a+1)}}, \tag{6.14}$$

$$\int_0^1 \frac{\log\Gamma(a+1+u) - \log\Gamma(a+1-u)}{\sqrt{u(1-u)}}\,\mathrm{d}u = 2\pi\log\frac{\sqrt{a}+\sqrt{a+1}}{2}. \tag{6.15}$$

To prove these, replace u by $1-u$ in the second member of each difference inside its integral. The recurrences for ψ and Γ then leave respectively $1/(a+u)$ and $\log(a+u)$. The first arcsine integral is obtained by $u = (1+\cos\theta)/2$; integrating it in a and comparing as $a \to \infty$ gives the second. This proof fixes the normalization of the logarithm, including the factor 2 in its denominator.

From $\gamma_m(a+1) - \gamma_m(a) = -\log^m(a)/a$ one likewise obtains the all-order identity

$$\int_0^1 \frac{\gamma_m(a+1+u) - \gamma_m(a+1-u)}{\sqrt{u(1-u)}}\,\mathrm{d}u = -\int_0^1 \frac{\log^m(a+u)}{(a+u)\sqrt{u(1-u)}}\,\mathrm{d}u. \tag{6.16}$$

For instance, its $m = 1$ value is

$$-\frac{\pi}{\sqrt{a(a+1)}}\log\frac{4a(a+1)}{(\sqrt{a}+\sqrt{a+1})^2}. \tag{6.17}$$

For a direct evaluation, set $A = a + \frac{1}{2}$, $B = \frac{1}{2}$, $D = \sqrt{A^2 - B^2}$ and $r = B/(A + D)$. Then $A + B\cos\theta = (A + D)(1 + 2r\cos\theta + r^2)/2$. The Fourier series of its logarithm and the Poisson kernel of its reciprocal give

$$\int_0^\pi \frac{\log(A + B\cos\theta)}{A + B\cos\theta}\,\mathrm{d}\theta = \frac{\pi}{D}\log\frac{2D^2}{A + D}.$$

Their series converge absolutely since $0 < r < 1$; substitution is (6.17) with the required negative sign.

There is also a canonical primitive without an arbitrary periodic ambiguity. Define $S_m(a) = \int_1^a \gamma_m(x)\,\mathrm{d}x$ for $a > 0$. The elementary integral

$$\int_1^2 \zeta(1 + v, a)\,\mathrm{d}a = \frac{\zeta(v, 1) - \zeta(v, 2)}{v} = \frac{1}{v}$$

shows, coefficient by coefficient, that $\int_1^2 \gamma_m(a)\,\mathrm{d}a = 0$. Integrating the shift recurrence consequently gives

$$S_m(a + 1) - S_m(a) = -\frac{\log^{m+1} a}{m + 1}. \tag{6.18}$$

The same symmetry argument yields

$$\int_0^1 \frac{S_m(a + 1 + u) - S_m(a + 1 - u)}{\sqrt{u(1 - u)}}\,\mathrm{d}u = -\frac{1}{m + 1}\int_0^1 \frac{\log^{m+1}(a + u)}{\sqrt{u(1 - u)}}\,\mathrm{d}u. \tag{6.19}$$

For $m = 0$, $S_0(a) = -\log\Gamma(a)$, and this reduces exactly to (6.15). Equations (6.11) and (6.19) exhibit two different sources of all-order identities: cancellation of a zeta pole in a beta transform, and integration of the elementary shift recurrence.

7 All-rank localization of ordered finite-difference sums

The incoming finite-difference report proves determinant identities for symmetrized assignments of orders and shifts, and explicitly asks which coordinates are needed for an individual ordered assignment [4, Sections 3 and 9]. We give a finite construction for arbitrary rank. Its central fact is a support cancellation: after the differences are transferred to the ordered Vandermonde polynomial, two isolated free labels annihilate the coefficient. The construction retains the individual assignment of every order and shift.

7.1 The ordered sum and its exact transported coefficient

Fix an integer $r \geq 2$ and put $d = 2r - 2$. For $\mathbf{s} = (s_1, \ldots, s_r) \in \mathbb{C}^r$ and $\operatorname{Re} a_i > 0$, define

$$w_i(n) = \frac{1}{d!}\sum_{j=0}^{d}(-1)^j\binom{d}{j}(n + a_i + j)^{-s_i}, \tag{7.1}$$

$$\mathcal{Z}_r(\mathbf{s}; \mathbf{a}) = \sum_{0 \leq n_1 < \cdots < n_r} \Delta(\mathbf{n})^2 \prod_{i=1}^{r} w_i(n_i), \tag{7.2}$$

where $\Delta(\mathbf{x}) = \prod_{i<v}(x_v - x_i)$. All powers use the principal logarithm in the right half-plane. We shall distinguish two domains:

$$\Omega = \{\operatorname{Re} s_i > 1,\ \operatorname{Re} a_i > 0 \text{ for every } i\}, \qquad \Omega_0 = \{\operatorname{Re} s_i > d + 1,\ \operatorname{Re} a_i > 0 \text{ for every } i\}.$$

Theorem 7.1 (Finite ordered reduction in every rank)**.** *The series (7.2) converges absolutely and locally normally on Ω. On Ω_0 it is an explicit finite sum of expressions*

$$\sum_{0\le N_1<\cdots<N_q} Q(N_1,\dots,N_q)\prod_{\ell=1}^{q}\prod_{i\in B_\ell}(N_\ell+A_i)^{-s_i}, \qquad \operatorname{Re} A_i>0, \tag{7.3}$$

multiplied, where necessary, by finite products of shifted powers. Here Q is a polynomial, the label sets B_ℓ are consecutive blocks, and

$$q\le\left\lfloor\frac{r+1}{2}\right\rfloor.$$

The contribution anchored at the lower boundary has at most $\lfloor r/2\rfloor$ free indices. Equations (7.13), (7.16), and (7.18) below give all coefficients and shifts by finite sums. Every expression in this construction converges absolutely on Ω_0. Their collective sum has a holomorphic continuation to Ω, equal there to (7.2).

The number q counts summation variables in the displayed representation. The theorem makes no assertion that this number is minimal, or that any special values obtained from different sectors are arithmetically independent.

For the convergence statement, the finite-difference integral is

$$w_i(n)=\frac{(s_i)_d}{d!}\int_{[0,1]^d}(n+a_i+u_1+\cdots+u_d)^{-s_i-d}\,\mathrm{d}\mathbf{u}. \tag{7.4}$$

The sign is positive because d is even. On compact subsets of the parameter domain this gives $|w_i(n)|\le C(1+n)^{-\operatorname{Re}s_i-d}$. For nonnegative integer nodes,

$$|\Delta(\mathbf{n})|^2\le\prod_{i=1}^{r}(1+n_i)^d. \tag{7.5}$$

The resulting product majorant proves normal convergence on Ω. If the differences in (7.1) are expanded before summation, each factor is only bounded by $C(1+n_i)^{-\operatorname{Re}s_i}$; the same argument then proves absolute convergence on Ω_0. This is the initial domain for the coefficient transport.

On integer tuples define the chamber polynomial

$$P(\mathbf{x})=\mathbf{1}_{\{0\le x_1<\cdots<x_r\}}\Delta(\mathbf{x})^2 \tag{7.6}$$

and the finite stencil

$$C_r(\mathbf{k})=\sum_{\mathbf{j}\in\{0,\dots,d\}^r}(-1)^{j_1+\cdots+j_r}\prod_{i=1}^{r}\binom{d}{j_i}P(\mathbf{k}-\mathbf{j}). \tag{7.7}$$

Expanding (7.1), setting $k_i=n_i+j_i$, and collecting the common shifted powers gives the exact identity

$$\mathcal{Z}_r(\mathbf{s};\mathbf{a})=\frac{1}{(d!)^r}\sum_{k_1,\dots,k_r\ge0}C_r(\mathbf{k})\prod_{i=1}^{r}(k_i+a_i)^{-s_i} \qquad \text{on } \Omega_0. \tag{7.8}$$

Expanded absolute convergence justifies every regrouping. There is no unaccounted upper endpoint: the regrouping is performed on the full absolutely convergent series. Also, $C_r(\mathbf{k})=0$ if any $k_i<0$, since every nonzero term in (7.7) has $k_i=x_i+j_i\ge0$.

7.2 Collision blocks and the two-singleton cancellation

The shifted tuple $\mathbf{k}$ need not be increasing. If $C_r(\mathbf{k}) \neq 0$, however, some stencil term has a strictly increasing tuple $\mathbf{k} - \mathbf{j}$. Necessarily

$$k_{i+1} - k_i \geq 1 - d \qquad (1 \leq i < r). \tag{7.9}$$

Partition the labels $1, \ldots, r$ into maximal consecutive blocks by placing a cut precisely when

$$k_{i+1} - k_i > d. \tag{7.10}$$

The internal adjacent gaps therefore belong to the finite set $\{1 - d, \ldots, d\}$. Across a cut, every stencil assignment remains strictly ordered, because $k_{i+1} - j_{i+1} - k_i + j_i \geq d + 1 - d = 1$.

The first block is called *anchored* when $k_1 < d$; every other block is free. If $k_1 \geq d$, all blocks are free. Only the first block can be anchored. Indeed, the chamber condition in (7.6) requires just $x_1 \geq 0$ together with strict order, which already implies nonnegativity of the later coordinates.

Lemma 7.2 (Two free singleton blocks give zero)**.** *If the block decomposition of* $\mathbf{k}$ *contains two free blocks of size one, then* $C_r(\mathbf{k}) = 0$.

Proof. Let i be one free singleton, and fix every stencil variable except j_i. All chamber indicators are independent of j_i: its neighboring cuts remain ordered for every shift, and it cannot change the lower boundary condition. As a polynomial in $x_i = k_i - j_i$, the Vandermonde square has degree $d = 2r - 2$, with leading coefficient

$$\Delta(\mathbf{x}_{\widehat{i}})^2 = \prod_{\substack{p<v \\ p,v\neq i}} (x_v - x_p)^2.$$

Consequently

$$\sum_{j_i=0}^{d} (-1)^{j_i} \binom{d}{j_i} \Delta(\mathbf{x})^2 = d!\, \Delta(\mathbf{x}_{\widehat{i}})^2.$$

Let v be the second free singleton. Its indicators are still independent of j_v, but the remaining polynomial has degree $2(r-2) = d - 2$ in x_v. Its dth backward difference is zero. Summing over the other stencil variables preserves this cancellation. □

At most one surviving free block can therefore be a singleton. If there are q free blocks and no anchor, they use at least $2q - 1$ labels; hence $q \leq \lfloor (r+1)/2 \rfloor$. If an anchor is present, it uses at least one additional label, giving $r \geq 2q$ and $q \leq \lfloor r/2 \rfloor$. This proves the support part of Theorem 7.1. We now turn it into a finite formula without leaving any block-coordinate choices implicit.

7.3 An explicit finite construction of every sector

Choose a consecutive block composition $\mathcal{B} = (B_1, \ldots, B_p)$ of $\{1, \ldots, r\}$, and write $b_\ell = |B_\ell|$ and $\ell(i)$ for the block containing i. In each block choose integer offsets u_i with first offset zero and

$$u_{i+1} - u_i \in \{1 - d, \ldots, d\} \quad (i, i+1 \in B_\ell). \tag{7.11}$$

There are finitely many choices. Put

$$k_i = t_{\ell(i)} + u_i, \qquad c_\ell = d + u_{\max B_\ell},$$

and introduce the finite internal indicator

$$\varepsilon_{\mathcal{B},\mathbf{u}}(\mathbf{j}) = \prod_{\ell=1}^{p} \prod_{i,i+1\in B_\ell} \mathbf{1}_{\{u_{i+1}-u_i+j_i-j_{i+1}\geq 1\}}.$$

Discard a pattern if this indicator is zero for every stencil tuple. For every retained pattern, an admissible tuple gives

$$u_i \geq i - \min B_\ell - d \quad (i \in B_\ell), \qquad c_\ell \geq b_\ell - 1 \geq 0. \tag{7.12}$$

To see the first inequality, sum the internal strict inequalities from the first label of the block to i, then use $j_i - j_{\min B_\ell} \geq -d$. The second is its instance at the last label.

Define the following polynomial in the block bases:

$$\mathcal{C}_{\mathcal{B},\mathbf{u}}(\mathbf{t}) = \sum_{\mathbf{j}\in\{0,\dots,d\}^r} (-1)^{|\mathbf{j}|} \prod_{i=1}^{r} \binom{d}{j_i} \varepsilon_{\mathcal{B},\mathbf{u}}(\mathbf{j}) \prod_{i<v} (t_{\ell(v)} - t_{\ell(i)} + u_v - u_i - j_v + j_i)^2. \tag{7.13}$$

It has integer coefficients and satisfies

$$\deg_{t_\ell} \mathcal{C}_{\mathcal{B},\mathbf{u}} \leq 2b_\ell(r - b_\ell). \tag{7.14}$$

Pairs of labels in the same block do not depend on its base; each of the $b_\ell(r - b_\ell)$ cross-block pairs contributes at most degree two.

For an unanchored sector the base conditions are exactly

$$t_1 \geq d, \qquad t_{\ell+1} > t_\ell + c_\ell.$$

Set

$$e_1 = d, \qquad e_\ell = d + \sum_{v<\ell} c_v, \qquad t_\ell = N_\ell + e_\ell. \tag{7.15}$$

The conditions become $0 \leq N_1 < \cdots < N_p$. Moreover, $e_\ell + u_i \geq 0$ by (7.12), so every shifted argument stays in the right half-plane. This sector contributes exactly

$$\frac{1}{(d!)^r} \sum_{0\leq N_1<\cdots<N_p} \mathcal{C}_{\mathcal{B},\mathbf{u}}(\mathbf{N}+\mathbf{e}) \prod_{\ell=1}^{p} \prod_{i\in B_\ell} (N_\ell + e_\ell + u_i + a_i)^{-s_i}. \tag{7.16}$$

The stencil polynomial is exactly $C_r(\mathbf{k})$ on this sector: cross-block order and the lower-bound condition are automatic, leaving precisely the indicators in (7.13).

For an anchored sector choose the finite integer $h = k_1 \in \{0, \dots, d-1\}$ and set $t_1 = h$. Let $\mathcal{C}_{\mathcal{B},\mathbf{u};h}(t_2, \dots, t_p)$ be (7.13) with $t_1 = h$ and the additional indicator $\mathbf{1}_{\{j_1\leq h\}}$ inserted in the summand. Discard patterns with no admissible stencil tuple under this extra constraint. If $p \geq 2$, set

$$e_2 = h + c_1 + 1, \qquad e_\ell = e_2 + \sum_{v=2}^{\ell-1} c_v \quad (\ell \geq 3), \qquad t_\ell = N_\ell + e_\ell. \tag{7.17}$$

The free variables satisfy $0 \leq N_2 < \cdots < N_p$. Feasibility of the anchored block gives $h + u_i \geq 0$ for its labels, and in particular $e_2 = d + h + u_{\max B_1} + 1 \geq d + 1$. Together with (7.12), this also ensures positive real parts for all the free shifted arguments. The anchored sector is

$$\frac{1}{(d!)^r} \prod_{i\in B_1} (h + u_i + a_i)^{-s_i} \sum_{0\leq N_2<\cdots<N_p} \mathcal{C}_{\mathcal{B},\mathbf{u};h}(N_2 + e_2, \dots, N_p + e_p) \times \prod_{\ell=2}^{p} \prod_{i\in B_\ell} (N_\ell + e_\ell + u_i + a_i)^{-s_i}. \tag{7.18}$$

For $p = 1$, the sum and the free product are absent, so the contribution is a finite scalar expression.

Every possible nonzero coefficient in (7.8) belongs to exactly one of these sectors: its cuts determine the composition, the first coordinate in each block determines the base, and subtraction determines the offsets. Conversely, the indicated inequalities give exactly those cuts and internal gaps. There are finitely many compositions, offsets, and anchored values h. Summing (7.16) and (7.18) over them, and discarding the zero sectors of Lemma 7.2, is therefore the promised finite construction.

For completeness, each separated sector converges absolutely on Ω_0. If a free block has b labels, its polynomial degree in the associated variable is at most $2b(r-b)$, while the sum of the real parts of its denominator exponents is greater than $b(d+1)$. Their difference satisfies

$$b(d+1) - 2b(r-b) = b(2b-1) \geq 1,$$

with strict improvement supplied by $\operatorname{Re} s_i > d+1$. A product majorant over unrestricted nonnegative free variables is summable. The same bounds apply locally uniformly to the parameters. This proves the convergence assertions and completes the proof of Theorem 7.1.

If a representation without polynomial numerators is preferred, choose one denominator $(N+A)^{-s}$ in each free block. Expanding $N^k = ((N+A)-A)^k$ removes any monomial numerator by finitely many integer decrements of that exponent. The preceding degree estimate proves absolute convergence of these terms on Ω_0 as well, and no new summation variable is introduced.

Remark 7.3 (Collective continuation and separated convergence). The original series is holomorphic on Ω. Since the finite localized expression agrees with it on the nonempty subdomain Ω_0, the former gives a holomorphic continuation of the latter collectively. The degree estimate above does not prove that each separated sector, or each individual exponent-shifted term, converges on all of Ω. Finite cancellation between offset patterns may be needed there. This distinction is part of the theorem's domain statement, not an appeal to formal rearrangement outside convergence.

7.4 Rank three: the complete two-index contact part

For $r = 3$, $d = 4$, write $f_i(k) = (k+a_i)^{-s_i}$. The unanchored composition $(1,2) \mid (3)$ is described by

$$k_1 \geq 4, \qquad h = k_2 - k_1 \in \{-3, \dots, 4\}, \qquad k_3 - k_2 > 4.$$

Summing the isolated third stencil first, exactly as in Lemma 7.2, gives

$$C_3(k_1,k_2,k_3) = 24A(h), \tag{7.19}$$

$$A(h) = \sum_{j,k=0}^{4} (-1)^{j+k} \binom{4}{j}\binom{4}{k} (h+j-k)^2 \mathbf{1}_{\{h+j-k>0\}}. \tag{7.20}$$

This entire integer contact table has a short generating expression:

$$\sum_{h\in\mathbb{Z}} A(h) z^h = (1-z^{-1})^4 (1-z)^4 \frac{z(1+z)}{(1-z)^3} = z^{-3}(1+z)(1-z)^5. \tag{7.21}$$

Indeed, $z(1+z)/(1-z)^3 = \sum_{h\geq 1} h^2 z^h$, and the two finite stencils multiply this series by the displayed Laurent factors. One may first work in $0 < |z| < 1$, where every intermediate series converges. The final Laurent polynomial proves

h	-3	-2	-1	0	1	2	3
$A(h)$	1	-4	5	0	-5	4	-1

$$A(h) = 0 \quad (|h| \geq 4).$$

The resulting two-index contribution of $(1,2) \mid (3)$ is

$$\mathcal{T}_{12|3} = \frac{1}{24^2} \sum_{h=-3}^{3} A(h) \sum_{0 \le n < m} f_1(n+4) f_2(n+4+h) f_3(m+8+h). \tag{7.22}$$

The other unanchored two-block composition, $(1) \mid (2,3)$, gives

$$\mathcal{T}_{1|23} = \frac{1}{24^2} \sum_{h=-3}^{3} A(h) \sum_{0 \le n < m} f_1(n+4) f_2(m+8) f_3(m+8+h). \tag{7.23}$$

Every remaining surviving sector has at most one free index. Thus

$$\mathcal{Z}_3 = \mathcal{T}_{12|3} + \mathcal{T}_{1|23} + \mathcal{E}_3, \tag{7.24}$$

where $\mathcal{E}_3$ is exactly the finite sum prescribed by (7.16)–(7.18) over the remaining compositions and anchor choices. More explicitly, it consists of the unanchored one-block sectors, all anchored one-block sectors, and the anchored two-block sectors; anchored three-block sectors and unanchored three-block sectors vanish by Lemma 7.2. This is a finite formula for its one-index correction, with no undetermined coefficient.

The contact contribution is not vacuous: the admissible tuple $(5,6,20)$ has $h = 1$ and

$$C_3(5,6,20) = 24A(1) = -120. \tag{7.25}$$

This proves the existence of two-index support in the construction. Further identities can still simplify the complete sum for special choices of the orders or shifts; (7.25) is not a minimal-depth assertion.

7.5 General integer stencils and stronger annihilation

The support mechanism depends on polynomial moments of the stencil, rather than on consecutive binomial coefficients. The next result retains $d = 2r - 2$, even when the stencil has a larger support.

Corollary 7.4 (Finite integer stencils)**.** *Let $L \ge d$ be an integer and, for each label i, choose coefficients $\alpha_{i,\lambda} \in \mathbb{C}$ supported on $0 \le \lambda \le L$ such that*

$$\sum_{\lambda=0}^{L} \alpha_{i,\lambda} \lambda^k = 0 \qquad (0 \le k < d). \tag{7.26}$$

Replace (7.1) by

$$w_{i,\alpha}(n) = \sum_{\lambda=0}^{L} \alpha_{i,\lambda} (n + a_i + \lambda)^{-s_i}.$$

The corresponding ordered sum has the conclusions of Theorem 7.1, with the same bounds $\lfloor (r+1)/2 \rfloor$ and $\lfloor r/2 \rfloor$ on its free indices. If every stencil also annihilates degree d, the bounds improve to $\lfloor r/2 \rfloor$ for the unanchored part and $\lfloor (r-1)/2 \rfloor$ for the anchored part.

Proof. The exact transported coefficient is now

$$C_\alpha(\mathbf{k}) = \sum_{\boldsymbol{\lambda} \in \{0,\dots,L\}^r} \left(\prod_i \alpha_{i,\lambda_i} \right) P(\mathbf{k} - \boldsymbol{\lambda}). \tag{7.27}$$

The factor $(d!)^{-r}$ is absent unless it was included in the chosen coefficients. Expanded absolute convergence on Ω_0 justifies the same transport. A nonzero coefficient has adjacent gaps at least $1 - L$. Make a cut at gaps greater than L, and anchor the first block when $k_1 < L$.

At a free singleton i, a degree-d polynomial is reduced to its leading coefficient times

$$\mu_i = \sum_{\lambda=0}^{L} \alpha_{i,\lambda}(-\lambda)^d.$$

This follows by expanding each power of $k_i - \lambda$ and using (7.26). A second free singleton then occurs in a polynomial of degree $d-2$, so its stencil annihilates the term. The previous counting argument gives the original bounds. If each $\mu_i = 0$, even the first free singleton annihilates the coefficient. Every free block then uses at least two labels; with or without an anchor this gives exactly the two improved bounds.

The full finite sector construction is unchanged after replacing the geometric thresholds d by L, the shift range by $\{0, \dots, L\}$, and the binomial stencil product by $\prod_i \alpha_{i,\lambda_i}$. In particular internal offset increments belong to $\{1-L, \dots, L\}$, $c_\ell = L + u_{\max B_\ell} \geq b_\ell - 1$, and the unanchored first base is L. The anchored first base has finitely many choices $0 \leq h < L$, with the condition $\lambda_1 \leq h$. Feasibility gives $u_i \geq i - \min B_\ell - L$, so the same base substitutions keep all shifted arguments in the right half-plane. The polynomial degree bound (7.14) does not change. Consequently Ω_0, with $d = 2r-2$, still suffices for convergence of every separated sector, independently of L.

Finally Taylor's formula with a uniform remainder, applied to $(n + a_i + \lambda)^{-s_i}$ through degree $d-1$ in λ, gives

$$w_{i,\alpha}(n) = O\big((1+n)^{-\operatorname{Re} s_i - d}\big)$$

locally uniformly in the parameters. The polynomial terms vanish by (7.26); the finite support bounds the remainders. Equation (7.5) proves original normal convergence on Ω, and the collective continuation follows as before. □

Every consecutive finite difference of order strictly greater than $2r-2$ satisfies the stronger hypothesis. For nonintegral rational stencils, transport onto one integer lattice would instead require separate residue classes; global moment cancellation alone does not justify the integer-sector argument above.

The transport itself is discrete summation by parts. Discrete residues provide a classical framework for rational summability [14]. Here the specific additional ingredient is the support of the ordered chamber coefficient and the loss of two Vandermonde degrees after a singleton is removed. That mechanism works with arbitrary complex orders and unequal shifts on the stated genuine convergence domains, and resolves the finite-coordinate representation question for this family in [4].

8 A Gamma integral basis and eventual reconstruction at every fixed gap

The preceding sections develop identities by localizing a finite-difference operator at its boundary. We now apply a different analytic construction to a remaining reconstruction problem in the repository: the optimal initial moments of two unequal even harmonic tails. The inherited fixed-gap reduction expresses the obstruction as a rational determinant of size half the gap [5]. We construct its entire null space by ordinary Gamma integrals, determine the exact normalization, and obtain a nonzero determinant asymptotic. Consequently the optimal prefix holds eventually on *every fixed-gap ray*. This differs from the previously established eventual result with the smaller order fixed [8]: here both orders tend to infinity while their difference remains fixed.

The Gamma recurrence, Stirling expansion, and Fourier-transform setting are classical [11, 12, 18]. The contributions established below are the explicit basis for this particular null

equation, its exact bridge to the inherited rational determinant, the fixed-gap asymptotic, and its harmonic-number correction. This is a statement of the advance relative to the inspected material; it does not assert historical priority. Throughout the section, rank means rank of the specified rational *formal coordinate matrix*. No arithmetic independence of evaluated zeta values or moments is assumed or concluded.

8.1 The inherited moments and the determinant to be studied

For integers $1 \le a < b$, write

$$p = 2a, \qquad q = 2b, \qquad m = b - a \ge 1, \qquad x_n = n + \tfrac{1}{2}.$$

Thus b is half the larger order and m is half the gap. Define

$$A_p(n) = \zeta(p, n+1) = \zeta(p) - H_n^{(p)}, \qquad D_{p,q}(s) = \sum_{n=0}^{\infty} A_p(n) A_q(n) x_n^{-s}. \tag{8.1}$$

The defining series converges absolutely for $\operatorname{Re} s > 3 - p - q$. Let $D_r = D_{p,q}(-2r)$, using meromorphic continuation when necessary. These points are regular. Indeed the centered Hurwitz expansion has the form

$$A_p(n) \sim \frac{x_n^{1-p}}{p-1} + \sum_{k \ge 1} \frac{B_{2k}(1/2)}{(2k)!} (p)_{2k-1} x_n^{1-p-2k}.$$

Subtracting finitely many terms from the product continues (8.1) successively to larger half-planes. Its possible poles lie at $s = 3 - p - q - 2\ell$, $\ell \ge 0$, all odd integers. In particular the even arguments used in D_r are regular.

The inherited two-tail calculus supplies blocks C_j and the mutually inverse Bernoulli transforms

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) C_j, \tag{8.2}$$

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r. \tag{8.3}$$

Here $B_n(x)$ denotes a Bernoulli polynomial and $B_n = B_n(0)$, with $B_1 = -1/2$. For $j \ge b$ the late coordinates are

$$C_j = \sum_{k=0}^{b-1} a_{jk} Z_k, \qquad Z_k = \zeta(2k), \qquad Z_0 = -\tfrac{1}{2}. \tag{8.4}$$

Their exact rational coefficients, also inherited, are as follows. Put $d = 2j+1$ and $e = d - p - q + 2k + 1$. If $e < 0$ set $a_{jk} = 0$; otherwise set

$$a_{jk} = B_e \left\{ \frac{\binom{d-q+1}{p-2k}}{d-q+1} + \frac{\binom{d-p+1}{q-2k}}{d-p+1} \right\}. \tag{8.5}$$

Each binomial summand is omitted unless its lower index lies between 2 and its upper index. In particular, a lower index of zero is *omitted*. This convention is essential: it determines the constant subtraction in the differential equation below. Equations (8.2)–(8.5) are inputs from [5, Section 2, inherited moment coordinates].

Define the $b \times b$ late-block matrix

$$M_{a,b} = (a_{b+r,k})_{0 \le r,k < b}. \tag{8.6}$$

On the ordered formal labels

$$C_0, \dots, C_{b-1}, \quad Z_0, \dots, Z_{b-1},$$

the coefficient matrix of $C_0, \dots, C_{2b-1}$ is $\operatorname{diag}(I_b, M_{a,b})$. The forward transform (8.2) is triangular with diagonal $1/(2r+1)$. Consequently the determinant of the complete matrix for the first $q = 2b$ moments $D_0, \dots, D_{2b-1}$ is

$$\frac{\det M_{a,b}}{(4b-1)!!}. \tag{8.7}$$

Thus the optimal-prefix question is exactly the nonvanishing of (8.6). If it is nonzero, those first $2b$ moments generate every later moment in the inherited formal coordinate system, and evaluation gives the corresponding rational generation statements for the actual constants. Fewer than $2b$ rows cannot have rank $2b$ on these $2b$ specified labels.

8.2 The fixed-gap recurrence and its exact normalization

For the remainder of the section fix $m \ge 1$. To distinguish the recurrence coefficients from both the tails and Bernoulli numbers, put

$$\mathfrak{a}_r = \begin{cases} 0, & 1 \le r \le m, \\ (2r)^{\underline{2m}}, & r > m, \end{cases} \qquad \mathfrak{b}_N = (2N-1)^{\underline{2m}}, \qquad d_N = 2(\mathfrak{a}_N + \mathfrak{b}_N) \quad (N > m). \tag{8.8}$$

The underline denotes a falling factorial. All d_N in this range are strictly positive. Let

$$\beta_k = \frac{2|B_{2k}|}{(2k)!} = \frac{4\zeta(2k)}{(2\pi)^{2k}} > 0, \qquad k \ge 1. \tag{8.9}$$

The rational fundamental array is defined by

$$C^{[m]}_{r,j} = \delta_{rj} \qquad (1 \le r, j \le m), \tag{8.10}$$

$$C^{[m]}_{N,j} = \frac{1}{d_N} \sum_{r=1}^{N-1} \beta_{N-r}(\mathfrak{a}_r + \mathfrak{b}_N) C^{[m]}_{r,j} \qquad (N > m,\ 1 \le j \le m). \tag{8.11}$$

The superscript separates this array from the moment blocks C_j. For $b > m$ write

$$F_{b,m} = (C^{[m]}_{b+i,j})_{1 \le i,j \le m}. \tag{8.12}$$

For an even formal series $R(z) = \sum_{n \ge 1} r_n z^{2n}$, define

$$\mathcal{T}_m R = \left(R^{(2m)} - R^{(2m)}(0)\right) \cot(z/2) + \left(R \cot(z/2)\right)^{(2m)}. \tag{8.13}$$

Both terms combine to an odd formal Taylor series: the subtraction removes the possible pole in the first product, while $R \cot(z/2)$ is an odd Taylor series. Expanding

$$\cot(z/2) = \frac{2}{z} - \sum_{k \ge 1} \beta_k z^{2k-1}$$

shows that, for $N > m$, its coefficient at $z^{2N-2m-1}$ is

$$d_N r_N - \sum_{r=1}^{N-1} \beta_{N-r}(\mathfrak{a}_r + \mathfrak{b}_N) r_r. \tag{8.14}$$

The value $\mathfrak{a}_m = 0$ records precisely the subtracted $R^{(2m)}(0)$. Since $d_N > 0$, the first m coefficients are arbitrary and determine all later ones uniquely. The formal null space of $\mathcal{T}_m$ has dimension exactly m.

Proposition 8.1 (Inherited boundary reduction, with normalization)**.** *For $b > m \geq 1$,*

$$\boxed{\det M_{b-m,b} = \frac{(-1)^{b(b-1)/2}}{4^b b!} \left(\prod_{N=m+1}^{b+m} d_N \right) \det F_{b,m}.} \tag{8.15}$$

This is the complementary-minor theorem of [5, Theorem `gap:complementary`*].*

Proof. We reproduce the normalization argument so that the subsequent Gamma determinant can be checked against the original coordinates. Set $h(t) = \coth(t/2)$ and

$$\mathcal{L}_m Q = (Q^{(2m)} - Q^{(2m)}(0))h + (Qh)^{(2m)}.$$

Let $K_{m,b}$ map the coefficients $(q_1, \dots, q_b)$ of $Q(t) = \sum_{n=1}^{b} q_n t^{2n}$ to the coefficients of $t, t^3, \dots, t^{2b-1}$ in $\mathcal{L}_m Q$. Write $h(t) = \sum_{k \geq 0} h_k t^{2k-1}$, where $h_0 = 2$ and $h_k = 2B_{2k}/(2k)!$ for $k \geq 1$. In row r, indexed from zero, put $N = r + m + 1$. The entry in column n is

$$h_{N-n}(\mathfrak{a}_n + \mathfrak{b}_N) \quad (n \leq N),$$

and is zero for $n > N$. In (8.5), substitution $j = b + r$, $k = b - n$ gives Bernoulli index $2(N - n)$ and the upper/lower binomial pairs

$$(2r + 2,\ 2n - 2m), \qquad (2r + 2m + 2,\ 2n).$$

Multiplication by $2(2n)!/(2r+1)!$ therefore gives exactly the displayed operator entry, including the omission when $n \leq m$. If M^{rev} denotes column reversal, then

$$K_{m,b} = \mathrm{diag}_{0 \leq r < b}\left(\frac{2}{(2r+1)!} \right) M^{\mathrm{rev}}_{b-m,b}\, \mathrm{diag}_{1 \leq n \leq b}((2n)!),$$

so

$$\det K_{m,b} = (-1)^{b(b-1)/2} 4^b b! \det M_{b-m,b}. \tag{8.16}$$

The relation $h_k = (-1)^{k-1}\beta_k$ for $k \geq 1$ and (8.11) imply that

$$Q_j(t) = \sum_{n \geq 1} (-1)^{n-j} C^{[m]}_{n,j} t^{2n}, \qquad 1 \leq j \leq m,$$

are formal null solutions of $\mathcal{L}_m$, with initial $m \times m$ coefficient block equal to the identity. Replace the first m columns of $K_{m,b}$ by the corresponding linear combinations given by the truncations of these solutions through $n = b$. This changes no determinant. The first $b - m$ rows of those columns vanish. Their bottom m rows equal $-TQ_+$, where T is lower triangular with diagonal $d_{b+1}, \dots, d_{b+m}$ and

$$(Q_+)_{ij} = (-1)^{b+i-j} C^{[m]}_{b+i,j}.$$

Indeed each truncated null equation is the negative of its omitted future terms, and its newest future coefficient has factor d_{b+i}. Moving the first m columns to the end leaves the other diagonal block triangular with diagonal $d_{m+1}, \dots, d_b$. It follows that

$$\det K_{m,b} = (-1)^{m(b-m)+m} \left(\prod_{N=m+1}^{b+m} d_N \right) \det Q_+.$$

Factoring its row and column signs gives $\det Q_+ = (-1)^{mb} \det F_{b,m}$. The combined exponent $2mb - m(m-1)$ is even. Equation (8.16) now proves (8.15). □

8.3 An explicit Gamma integral basis

Define the positive even weight

$$W_m(x) = \left| \Gamma\left(\tfrac{1}{4} + \tfrac{ix}{2}\right) \right|^{4m} \qquad (x \in \mathbb{R}) \tag{8.17}$$

and the odd multipliers

$$P_j(x) = \sinh(\pi(m-j)x) \quad (1 \le j < m), \qquad P_m(x) = \tanh(\pi x). \tag{8.18}$$

Their order is chosen to agree with their distinct decay scales.

Theorem 8.2 (Gamma basis and exact top moments)**.** *For $1 \le j \le m$, the integral*

$$R_j(z) = 4 \sin(z/2) \int_0^\infty \sinh(xz) P_j(x) W_m(x) \, \mathrm{d}x \tag{8.19}$$

defines an even holomorphic function in $|\operatorname{Re} z| < j\pi$. Its germ at zero satisfies $\mathcal{T}_m R_j = 0$ and

$$R_j^{(2m)}(0) = 0 \quad (j < m), \qquad R_m^{(2m)}(0) = (-1)^{m+1} 2^{2m+2} \pi^m. \tag{8.20}$$

The m germs form a basis of the formal null space of (8.13). Their Taylor radii are exactly $j\pi$.

The contour argument below proves the null equation and the exact derivatives. The basis assertion will follow from the positive initial determinant in proposition 8.4; the exact Taylor radii will follow independently from proposition 8.6. This order of proof avoids assuming either linear independence or the location of a singularity from the integral representation alone.

Contour proof. Stirling's formula gives, as $x \to +\infty$,

$$W_m(x) = 2^m (2\pi)^{2m} x^{-m} e^{-m\pi x} (1 + O(x^{-2})).$$

Together with (8.18), this proves absolute convergence in the stated strips, locally uniform convergence of all differentiated integrals, and evenness of (8.19). We use the usual meromorphic Gamma function and the integer-power continuation

$$W_m(y) = \left[\Gamma(\tfrac{1}{4} + iy/2) \Gamma(\tfrac{1}{4} - iy/2) \right]^{2m}.$$

There is no branch choice for this continuation; the absolute-value notation in (8.17) is used only on the real line.

For any one of the allowed odd multipliers P, put

$$G_P(t) = \int_{\mathbb{R}} e^{ixt} P(x) W_m(x) \, \mathrm{d}x, \qquad Q_P(t) = 2 \sinh(t/2) G_P(t).$$

These integrals and all their derivatives converge absolutely for real t and define analytic germs. Since G_P is odd, Q_P is even and vanishes at zero. With $D = d/dt$, let

$$E_P(t) = e^t(D + \tfrac{1}{2})^{2m}G_P(t) - e^{-t}(D - \tfrac{1}{2})^{2m}G_P(t).$$

The identities $Q_P = (e^{t/2} - e^{-t/2})G_P$ and $Q_P \coth(t/2) = (e^{t/2} + e^{-t/2})G_P$ yield, by differentiation,

$$\coth(t/2)Q_P^{(2m)}(t) + (Q_P(t)\coth(t/2))^{(2m)} = \frac{E_P(t)}{\sinh(t/2)}. \tag{8.21}$$

In the first integral defining E_P, substitute $y = x - i$; its contour is $\mathbb{R} - i$. In the second substitute $y = x + i$; its contour is $\mathbb{R} + i$. The factors $e^{\pm t}$ disappear under these substitutions. Gamma recurrence gives exactly

$$(iy - \tfrac{1}{2})^{2m}W_m(y + i) = (iy + \tfrac{1}{2})^{2m}W_m(y - i) = 2^{2m}V_m(y), \tag{8.22}$$

where

$$V_m(y) = \left[\Gamma(\tfrac{3}{4} + iy/2)\Gamma(\tfrac{3}{4} - iy/2)\right]^{2m}.$$

For example the first equality shifts $\Gamma(-1/4 + iy/2)$ to $\Gamma(3/4 + iy/2)$; its recurrence factor is $iy/2 - 1/4$. The second equality has the opposite sign before taking the even power, hence the same factor 2^{2m}.

Every allowed multiplier satisfies $P(y+i) = P(y-i)$. Both shifted integrands are therefore the same meromorphic function

$$\mathcal{F}_P(y) = 2^{2m}e^{iyt}V_m(y)P(y + i).$$

The nearest Gamma poles of V_m are $y = \pm 3i/2$, outside $|\Im y| \le 1$. For $P(x) = \sinh(\pi k x)$, $1 \le k < m$, the multiplier is $P(y + i) = (-1)^k \sinh(\pi k y)$, an entire function. Uniform Stirling estimates on the closed strip give a polynomial times $e^{-\pi(m-k)|\operatorname{Re} y|}$ as a bound for the integrand, with a constant depending on the fixed real t. For the tanh multiplier, $P(y + i) = \tanh(\pi y)$; away from its finite poles it is bounded at the two ends of the strip, and the decay is a polynomial times $e^{-m\pi|\operatorname{Re} y|}$. Thus the vertical sides of the rectangle between the two horizontal contours vanish as their real parts tend to infinity. Its bottom side runs from left to right, so bottom minus top equals $+2\pi i$ times the enclosed residue sum.

In the sinh case there are no enclosed poles and $E_P(t) = 0$. In the tanh case the only enclosed poles are the simple poles $y = \pm i/2$. Each tanh residue is $1/\pi$, and $V_m(i/2) = V_m(-i/2) = \pi^m$. Hence

$$\operatorname{Res}_{y=\pm i/2}\mathcal{F}_P(y) = 2^{2m}\pi^{m-1}e^{\mp t/2},$$

and

$$E_{\tanh}(t) = i\,2^{2m+2}\pi^m \cosh(t/2). \tag{8.23}$$

All shifts have now been justified using pole-free Gamma factors, the two explicitly listed tanh poles, and decaying vertical sides.

The second summand on the left of (8.21) is analytic at zero. Its first summand has Laurent coefficient $2Q_P^{(2m)}(0)$ at t^{-1}. Comparing this coefficient shows that $Q_P^{(2m)}(0) = 0$ for a sinh multiplier and $Q_P^{(2m)}(0) = i2^{2m+2}\pi^m$ for tanh. Equations (8.21) and (8.23) therefore give the exact germ identity

$$(Q_P^{(2m)} - Q_P^{(2m)}(0))\coth(t/2) + (Q_P\coth(t/2))^{(2m)} = 0.$$

Finally set $R_P(z) = iQ_P(iz)$. Pairing positive and negative x in $G_P(iz)$ gives precisely (8.19). Rotation uses $\coth(iz/2) = -i\cot(z/2)$ and yields $\mathcal{T}_m R_P = 0$. Since $R_P^{(2m)}(0) = i(-1)^m Q_P^{(2m)}(0)$, it also gives (8.20). □

Corollary 8.3 (Exact ordinary Gamma moments)**.** *Put* $\phi_n(x) = \Im(x + i/2)^{2n}$. *The Taylor coefficients of* (8.19) *are*

$$t_{n,j} := [z^{2n}]R_j(z) = \frac{4}{(2n)!}\int_0^\infty \phi_n(x)P_j(x)W_m(x)\,\mathrm{d}x. \tag{8.24}$$

For every $m \geq 1$,

$$\int_0^\infty \phi_m(x)\sinh(\pi k x)W_m(x)\,\mathrm{d}x = 0 \qquad (1 \leq k < m), \tag{8.25}$$

$$\int_0^\infty \phi_m(x)\tanh(\pi x)W_m(x)\,\mathrm{d}x = (-1)^{m+1}2^{2m}\pi^m. \tag{8.26}$$

All these integrals converge absolutely.

Proof. The elementary generating identity

$$\sin(z/2)\sinh(xz) = \sum_{n\geq 1}\frac{\phi_n(x)}{(2n)!}z^{2n}$$

and local uniform exponential domination justify coefficient extraction. Now apply (8.20). □

The vanishing moments have a concrete polynomial interpretation. Since $\phi_m(x)/x$ has degree $2m-2$ and its positive zeros are

$$\frac{1}{2}\cot\frac{r\pi}{2m}, \qquad 1 \leq r < m,$$

equation (8.25) is an explicit family of multiple-orthogonality relations against positive weights on $(0,\infty)$. The zeros follow by writing $x + i/2 = |x + i/2|e^{i\theta}$, with $0 < \theta < \pi/2$, and solving $\sin(2m\theta) = 0$. These relations do not imply that every later consecutive coefficient determinant has one fixed sign.

8.4 A positive initial determinant and the exact Gamma representation

Write

$$T_m = (t_{n,j})_{1\leq n,j\leq m}, \qquad \Delta(y_1,\ldots,y_m) = \prod_{r<s}(y_s - y_r).$$

Proposition 8.4 (Positive normalization)**.** *The initial determinant is*

$$\det T_m = (-1)^{m(m-1)/2}K_m, \tag{8.27}$$

where $K_m > 0$ *has the absolutely convergent integral representation*

$$\begin{aligned} K_m = {} & \frac{2^{(m-1)(m-2)/2}}{m!}\left(\prod_{n=1}^{m}\frac{4n}{(2n)!}\right) \\ & \times \int_{(0,\infty)^m}\Delta(x_1^2,\ldots,x_m^2)\Delta(\cosh\pi x_1,\ldots,\cosh\pi x_m) \\ & \qquad \times \prod_{r=1}^{m} x_r W_m(x_r)\tanh(\pi x_r)\,\mathrm{d}x_1\cdots\mathrm{d}x_m. \end{aligned} \tag{8.28}$$

Consequently the R_j *form a basis of the full formal null space, and for every* $b > m$,

$$\boxed{\det F_{b,m} = \frac{(-1)^{m(m-1)/2}}{K_m}\det(t_{b+i,j})_{1\leq i,j\leq m}.} \tag{8.29}$$

Proof. The polynomial $\phi_n(x)$ has leading term nx^{2n-1} and only lower odd powers thereafter. A triangular row transformation therefore gives

$$\det T_m = \left(\prod_{n=1}^{m} \frac{4n}{(2n)!}\right) \det\left(\int_0^\infty x^{2n-1} P_j(x) W_m(x)\,\mathrm{d}x\right)_{n,j=1}^{m}.$$

The continuous Cauchy–Binet identity writes the remaining determinant as $1/m!$ times the integral of the two alternants $\det(x_r^{2n-1})_{n,r}$ and $\det(P_j(x_r))_{r,j}$ against $\prod_r W_m(x_r)\,\mathrm{d}x_r$. This identity follows directly by expanding the two determinants, applying Fubini, and permuting the integration variables; absolute convergence justifies each operation. The first alternant is $\prod_r x_r\,\Delta(x_1^2,\dots,x_m^2)$.

For the second alternant put $u = \pi x$, $v = \cosh u$, and use

$$\frac{\sinh(ku)}{\tanh u} = vU_{k-1}(v),$$

where U_{k-1} is the Chebyshev polynomial of the second kind, with degree $k-1$ and leading coefficient 2^{k-1}. After factoring $\tanh(\pi x_r)$ from row r, the column degrees are $m-1, m-2, \dots, 1, 0$. Their leading-coefficient product is $2^{(m-1)(m-2)/2}$. Reversing the columns thus gives

$$\begin{aligned}\det(P_j(x_r)) = (-1)^{m(m-1)/2} 2^{(m-1)(m-2)/2} \\ \times \prod_r \tanh(\pi x_r)\,\Delta(\cosh \pi x_1, \dots, \cosh \pi x_m).\end{aligned}$$

This proves (8.27)–(8.28). For distinct positive x_r, the two Vandermondes have the same sign, since both $x \mapsto x^2$ and $x \mapsto \cosh(\pi x)$ are strictly increasing. The integrand in (8.28) is strictly positive away from the collision hyperplanes. Its exponential growth from the cosh Vandermonde is at most $\prod_r e^{(m-1)\pi x_r}$, while each $W_m(x_r)$ supplies $e^{-m\pi x_r}$ times a polynomial factor. At zero all factors are bounded or vanish. This proves convergence and $K_m > 0$.

All R_j are null germs, and their first m coefficient columns are independent by (8.27). The formal null space has dimension m by (8.14), so they form a basis. Their coefficient vectors obey (8.11); consequently

$$(t_{b+i,j})_{i,j=1}^{m} = F_{b,m} T_m.$$

Taking determinants proves (8.29). □

8.5 A Laplace-moment lemma with controlled error

The asymptotic argument must retain one term beyond the leading Laplace phase. At every even mode that leading phase is real and its imaginary part vanishes. The following lemma supplies the necessary next term without requiring an analytic continuation of the error in a real density.

Lemma 8.5. *Let $m \geq 1$ be fixed, $R > 0$, and $a \in \mathbb{R} \setminus \{0\}$. Suppose a real, locally bounded function f on $[0,\infty)$ satisfies*

$$f(x) = ax^{-m} e^{-Rx}(1 + O(x^{-2})) \qquad (x \to \infty).$$

As integers $N \to \infty$,

$$\int_0^\infty (x + i/2)^N f(x)\,\mathrm{d}x = a e^{iR/2} \frac{\Gamma(N-m+1)}{R^{N-m+1}} \left(1 + \frac{imR}{2(N-m)} + O(N^{-2})\right). \tag{8.30}$$

The implicit constant may depend on f, m, R.

Proof. Choose fixed $X > 1$ beyond which the asserted density estimate holds. The integral over $[0, X]$ is $O(C^N)$ for a fixed C, hence negligible compared with $S_N = \Gamma(N - m + 1)R^{-N+m-1}$ by Stirling's formula. The absolute error made on $[X, \infty)$ by replacing $f(x)$ with $ax^{-m}e^{-Rx}$ is bounded by a constant times

$$\int_X^\infty |x + i/2|^N x^{-m-2} e^{-Rx}\, \mathrm{d}x.$$

Split this integral at $x = N^{2/3}$. On the lower part it is at most $\exp(\frac{2}{3}N \log N + O(N))$, whereas $S_N = \exp(N \log N + O(N))$. This part is smaller than any fixed inverse power of N times S_N. On the upper part,

$$\frac{|x + i/2|^N}{x^N} = \left(1 + \frac{1}{4x^2}\right)^{N/2} \le \exp\left(\frac{N}{8x^2}\right)$$

is uniformly bounded. Its integral is therefore at most a constant times $\Gamma(N-m-1)R^{-N+m+1} = O(N^{-2}S_N)$. This estimate uses only real-variable bounds on the density error.

For the model integral set $y = x + i/2$. After factoring $e^{iR/2}$, its integrand is $y^{N-m}(1 - i/(2y))^{-m}e^{-Ry}$ on the ray $[X, \infty) + i/2$. Shift that ray to $[X, \infty)$ inside $\operatorname{Re} y \ge X$. No pole is crossed. The short vertical segment at X contributes $O(C^N)$, and the vertical segment at infinity vanishes for each N because of $e^{-R \operatorname{Re} y}$. Uniformly for real $y \ge X$,

$$(1 - i/(2y))^{-m} = 1 + \frac{im}{2y} + O(y^{-2}).$$

Integration of the first two terms, extending their lower limit to zero at another $O(C^N)$ cost, gives

$$ae^{iR/2}\left\{\frac{\Gamma(N - m + 1)}{R^{N-m+1}} + \frac{im}{2}\frac{\Gamma(N - m)}{R^{N-m}}\right\}.$$

The last term has absolute size $O(N^{-2}S_N)$ by the same Gamma ratio estimate. This proves (8.30). □

Define the constants

$$\lambda_j = j\pi, \qquad \xi_j = (j\pi)^{-2}, \qquad A_m^* = 2^m(2\pi)^{2m}, \qquad a_j = \begin{cases} A_m^*/2, & j < m, \\ A_m^*, & j = m. \end{cases} \tag{8.31}$$

Conjugate Gamma asymptotics contain only even inverse powers after the leading factor. Thus

$$P_j(x)W_m(x) = a_j x^{-m} e^{-\lambda_j x}(1 + O(x^{-2})). \tag{8.32}$$

The further exponential terms in sinh or tanh are absorbed into the indicated error. In particular, there is no x^{-1} correction.

Proposition 8.6 (Distinct coefficient scales and exact radii)**.** *Put* $d_j = m + \mathbf{1}_{2|j}$ *and*

$$L_j = \begin{cases} 4a_j \sin(j\pi/2)\lambda_j^{m-1}, & j \text{ odd}, \\ 2ma_j \cos(j\pi/2)\lambda_j^m, & j \text{ even}. \end{cases} \tag{8.33}$$

All L_j *are nonzero, and*

$$t_{n,j} = L_j \lambda_j^{-2n}(2n)^{-d_j}(1 + O(n^{-1})). \tag{8.34}$$

The Taylor radius of R_j *is exactly* $\lambda_j = j\pi$.

Proof. Apply lemma 8.5 to (8.32), take the imaginary part, and multiply by $4/(2n)!$. If j is odd, $e^{i\lambda_j/2} = \pm i$ and the leading term survives. Since $\Gamma(2n-m+1)/\Gamma(2n+1) \sim (2n)^{-m}$, this gives the first line of (8.33). If j is even, the leading phase is ± 1 and its imaginary part is zero. The next term in (8.30) gives instead

$$\frac{4a_j \cos(j\pi/2)}{(2n)!} \frac{m}{2} \frac{\Gamma(2n-m)}{\lambda_j^{2n-m}},$$

which is the second line of (8.33) and the exponent $m+1$. The $O(N^{-2})$ error in the lemma is small enough to give a relative $O(n^{-1})$ error even in this second case. Finally (8.34) implies $\lim_{n\to\infty} |t_{n,j}|^{1/(2n)} = \lambda_j^{-1}$. The Cauchy–Hadamard formula proves the exact radius. This also provides another proof of independence: the smallest surviving radius in any nontrivial linear combination cannot cancel. □

8.6 Eventual optimal reconstruction for every fixed gap

Set

$$D_m = m^2 + \lfloor m/2 \rfloor, \qquad \mathcal{A}_m = \frac{\prod_{j=1}^m |L_j|}{K_m} \left(\prod_{j=1}^m \xi_j \right) \prod_{j<k} (\xi_j - \xi_k) > 0. \tag{8.35}$$

Theorem 8.7 (Fixed-gap determinant asymptotic)**.** *For each fixed integer $m \geq 1$, as $b \to \infty$,*

$$\boxed{\det F_{b,m} \sim (-1)^{\lfloor m^2/4 \rfloor} \mathcal{A}_m (2b)^{-D_m} \left[\pi^{2m} (m!)^2\right]^{-b}.} \tag{8.36}$$

In particular, there exists an integer $B(m) > m$ such that every pair $(p,q) = (2b-2m, 2b)$ with $b \geq B(m)$ has an optimal first-q formal moment prefix. Its late-block determinant has eventual sign

$$\operatorname{sgn} \det M_{b-m,b} = (-1)^{b(b-1)/2 + \lfloor m^2/4 \rfloor}. \tag{8.37}$$

Proof. From column j of $(t_{b+i,j})_{i,j=1}^m$, factor $L_j \lambda_j^{-2b} (2b)^{-d_j}$. By (8.34), the remaining matrix tends entrywise to $V = (\xi_j^i)_{i,j=1}^m$. Its determinant is

$$\det V = \left(\prod_j \xi_j \right) \prod_{j<k} (\xi_k - \xi_j) \neq 0.$$

The distinct numbers $\xi_1 > \cdots > \xi_m > 0$ give sign $(-1)^{m(m-1)/2}$, which cancels the same sign in (8.29). Furthermore

$$\prod_j \operatorname{sgn} L_j = (-1)^{\sum_{j=1}^m \lfloor j/2 \rfloor} = (-1)^{\lfloor m^2/4 \rfloor}, \qquad \sum_j d_j = m^2 + \lfloor m/2 \rfloor,$$

and $\prod_j \lambda_j^{-2b} = [\pi^{2m}(m!)^2]^{-b}$. These identities prove (8.36) with the positive constant (8.35). A sequence with a nonzero asymptotic equivalent is nonzero and has the same sign for all sufficiently large indices. Apply the exact positive-factor bridge (8.15) and then (8.7). □

Remark 8.8 (Scope of the theorem). The assertion is eventual for each *fixed gap* $2m$ and does not provide a numerical value of $B(m)$. It leaves only finitely many possible exceptional pairs on each such ray, but does not prove the entire gap-six family or the general all-unequal-even conjecture. No uniform assertion as m grows with b follows from the fixed-m errors used here.

For $m = 1$, (8.26) gives $K_1 = 8\pi$, and (8.35) gives $\mathcal{A}_1 = 4/\pi$. The fundamental adjacent-gap sequence therefore satisfies

$$C^{[1]}_{b+1,1} \sim \frac{2}{\pi b}\,\pi^{-2b}.$$

For $m = 2$ the eventual sign of $\det F_{b,2}$ is negative, consistent with the inherited theorem for every gap-four pair. For $m = 3$ it is positive. Finite exact checks give $\det F_{b,3} < 0$ at $b = 4, 5, 6$ and $\det F_{b,3} > 0$ at $7 \le b \le 40$. These observations illustrate a sign transition; the proof of eventual positivity is theorem 8.7, and the finite checks are not a proof of nonvanishing for every gap-six pair.

8.7 All-orders coefficients and a harmonic-number correction

The Gamma construction supplies more than a leading equivalent. It gives an explicit Poincare expansion to every finite order, and the first determinant correction has a concise harmonic-number form. Let Euler numbers be defined by

$$\operatorname{sech} z = \sum_{k\ge 0} E_{2k}\frac{z^{2k}}{(2k)!}.$$

Define rational numbers $w_{m,r}$ by the formal identity

$$\sum_{r\ge 0} w_{m,r}X^r = \exp\left(\sum_{k\ge 1}\frac{m|E_{2k}|}{k\,2^{2k+1}}X^k\right). \tag{8.38}$$

Thus $w_{m,0} = 1$ and $w_{m,1} = m/8$. For $\ell \ge 0$ set

$$b_{m,\ell} = \sum_{r=0}^{\lfloor \ell/2\rfloor} w_{m,r}\frac{(m+2r)_{\ell-2r}}{(\ell-2r)!}(i/2)^{\ell-2r}. \tag{8.39}$$

Proposition 8.9 (All-orders mode construction)**.** *For each fixed mode j, with λ_j, a_j as in (8.31),*

$$\int_0^\infty (x+i/2)^N P_j(x)W_m(x)\,\mathrm{d}x \sim a_j e^{i\lambda_j/2}\sum_{\ell\ge 0} b_{m,\ell}\frac{\Gamma(N-m-\ell+1)}{\lambda_j^{N-m-\ell+1}}. \tag{8.40}$$

This is a Poincare expansion: truncation after any fixed $\ell = L$ has error of size $O(\Gamma(N-m-L)\lambda_j^{-N+m+L})$. Taking imaginary parts retains even ℓ when j is odd and odd ℓ when j is even. In particular,

$$t_{n,j} = L_j\lambda_j^{-2n}(2n)^{-d_j}\left(1 + \frac{d_j(d_j-1)}{4n} + O(n^{-2})\right). \tag{8.41}$$

Proof. The shifted logarithmic Stirling expansion is

$$\log\Gamma(z+\alpha) \sim (z+\alpha-\tfrac12)\log z - z + \tfrac12\log(2\pi) + \sum_{n\ge 1}\frac{(-1)^{n+1}B_{n+1}(\alpha)}{n(n+1)z^n},$$

with a remainder of the next prescribed algebraic order in each closed subsector of $|\arg z| < \pi$ [12]. Apply it with $\alpha = 1/4$ and $z = \pm ix/2$. The conjugate pair cancels odd inverse powers. Using

$$B_{2k+1}(1/4) = -\frac{(2k+1)E_{2k}}{4^{2k+1}}$$

gives

$$\log W_m(x) \sim \log A_m^* - m\log x - m\pi x + \sum_{k\ge 1}\frac{m|E_{2k}|}{k\,2^{2k+1}x^{2k}}.$$

Exponentiation to any fixed order proves

$$W_m(x) \sim A_m^* x^{-m} e^{-m\pi x} \sum_{r\geq 0} w_{m,r} x^{-2r}. \tag{8.42}$$

The corrections to the leading exponential in P_j are themselves exponentially smaller, so they contribute less than any fixed inverse power after the leading density has been removed.

For each term $x^{-m-2r}e^{-\lambda_j x}$ in the resulting density, repeat the shift $y = x + i/2$ from lemma 8.5. Its algebraic factor becomes

$$y^{-m-2r}\left(1-\frac{i}{2y}\right)^{-m-2r} = y^{-m-2r}\sum_{h\geq 0}\frac{(m+2r)_h}{h!}\left(\frac{i}{2y}\right)^h.$$

Collecting terms with $2r+h=\ell$ gives exactly (8.39). To justify any fixed truncation, split at fixed $X>1$ as in the lemma, retain the finitely many necessary density terms, and bound its real remainder by the first omitted power of x^{-1} times the leading exponential. The split at $N^{2/3}$ bounds its moment by the corresponding Gamma ratio. The finitely many model contours have the same pole-free deformation, and Taylor's remainder for $(1-i/(2y))^{-m-2r}$ is uniform for $y \geq X$. These bounds prove the stated remainder in (8.40); no contour continuation of the real Stirling remainder is used.

For even ℓ the coefficient $b_{m,\ell}$ is real, and for odd ℓ it is purely imaginary. Thus the phase $e^{ij\pi/2}$ gives the asserted parity selection. At odd j the leading imaginary term has $\ell=0$ and the next possible one has $\ell=2$. At even j the leading one has $\ell=1$, with $b_{m,1}=im/2$, and the next possible one has $\ell=3$. In either case the first relative inverse-power correction comes solely from the factorial ratio

$$\frac{\Gamma(N-d+1)}{\Gamma(N+1)} = N^{-d}\left(1+\frac{d(d-1)}{2N}+O(N^{-2})\right) \qquad (d\in\mathbb{N}).$$

For integer d this follows directly by expanding $\prod_{r=0}^{d-1}(N-r)^{-1}$. Set $N=2n$ and $d=d_j$ to obtain (8.41). □

Theorem 8.10 (The harmonic first correction)**.** *Let $e=\lfloor m/2\rfloor$ and $H_0=0$. Define*

$$\boxed{\kappa_m = -\frac{m^2(m+3)+e(2m+3)}{4} + \sum_{r=1}^{e} r(H_{m+2r}-H_{m-2r}).} \tag{8.43}$$

Then

$$\begin{aligned}\det F_{b,m} = (-1)^{\lfloor m^2/4\rfloor}\mathcal{A}_m(2b)^{-D_m}\left[\pi^{2m}(m!)^2\right]^{-b} \\ \times\left(1+\frac{\kappa_m}{b}+O(b^{-2})\right).\end{aligned} \tag{8.44}$$

More generally, this determinant has an expansion to every finite inverse-power order. For fixed m, all its normalized coefficients belong to $\mathbb{Q}[\pi^2]$.

Proof. Factor the same column scales as in the proof of theorem 8.7. By (8.41), the remaining entry in row i, column j is

$$\xi_j^i\left\{1+\frac{1}{b}\left(\frac{d_j(d_j-1)}{4}-id_j\right)+O(b^{-2})\right\}.$$

For $V=(\xi_j^i)_{i,j=1}^m$, the first relative correction in its determinant is therefore

$$\frac{1}{4}\sum_{j=1}^m d_j(d_j-1) - \sum_{j=1}^m d_j\eta_j, \qquad \eta_j = \left(V^{-1}\operatorname{diag}(1,\dots,m)V\right)_{jj}. \tag{8.45}$$

The diagonal quantities can be computed without matrix inversion. Differentiating the Vandermonde product gives

$$\eta_j = \xi_j \frac{\partial}{\partial \xi_j} \log\det V = 1 + \sum_{k\neq j} \frac{\xi_j}{\xi_j - \xi_k}.$$

For $\xi_j = (j\pi)^{-2}$, partial fractions give

$$\eta_j = m + \frac{3}{4} + \frac{j}{2}(H_{m-j} - H_{m+j}). \tag{8.46}$$

Indeed $\xi_j/(\xi_j - \xi_k) = k^2/(k^2 - j^2)$, and

$$\sum_{k\neq j} \left(\frac{1}{k-j} - \frac{1}{k+j} \right) = H_{m-j} - H_{m+j} + \frac{3}{2j}.$$

Also $\sum_j \eta_j = m(m+1)/2$, either by the trace or by pairing the summands in the Vandermonde expression. Since $d_j = m + \mathbf{1}_{2|j}$, substitution into (8.45) gives

$$-\frac{m^2(m+3)}{4} + \frac{me}{2} - \sum_{r=1}^{e} \eta_{2r}.$$

Equation (8.46) now simplifies it to (8.43), proving (8.44).

For higher orders, insert any desired finite truncation of (8.40) into the fixed-size determinant. Its leading Vandermonde is nonzero, so division by that leading term gives a valid Poincare expansion. Relative to L_j, parity selection leaves only even powers of $\lambda_j = j\pi$ in the mode corrections. The factorial-ratio expansions have rational coefficients, while ratios of Vandermonde expressions in the ξ_j cancel their common powers of π^{-2}. Thus each normalized determinant coefficient is a polynomial in π^2 with rational coefficients, as claimed. □

For orientation, the first correction values are

$$\kappa_1 = -1, \qquad \kappa_2 = -\frac{14}{3}, \qquad \kappa_3 = -\frac{217}{15}, \qquad \kappa_4 = -\frac{949}{35}.$$

The series in (8.38) and (8.40) are formal asymptotic constructions; convergence is not asserted. The argument also does not establish a complete exponentially improved expansion or an explicit error constant sufficient to certify a numerical $B(m)$.

8.8 A hypergeometric form of the null equation

The differential equation itself has a useful classical hypergeometric form. Let Q be any even analytic null germ of $\mathcal{L}_m$ and write $c = Q^{(2m)}(0)$. On the local branches near $u = 1$ with $\log 1 = 0$ and $\sqrt{1} = 1$, define

$$Y(u) = \frac{Q(\frac{1}{2}\log u)}{\sqrt{u} - 1}, \qquad \Theta = u\frac{d}{du}.$$

The apparent singularity at $u = 1$ is removable because $Q(t) = O(t^2)$. Then

$$\left[\Theta^{2m} - u(\Theta + \tfrac{1}{2})^{2m}\right] Y(u) = -\frac{c}{2^{2m+1}}(1 + \sqrt{u}). \tag{8.47}$$

To verify it, set $Q(t) = (e^t - 1)F(t)$, so that $Q(t)\coth(t/2) = (e^t + 1)F(t)$. Multiplication of $\mathcal{L}_m Q = 0$ by $e^t - 1$ gives

$$D^{2m}F - e^{2t}(D+1)^{2m}F = -\tfrac{c}{2}(1 + e^t).$$

Now $u = e^{2t}$ and $D = 2\Theta$, proving (8.47). Its homogeneous operator is the standard generalized hypergeometric operator associated with

$$ {}_{2m}F_{2m-1}\left(\begin{matrix}\frac{1}{2},\dots,\frac{1}{2}\\ 1,\dots,1\end{matrix};u\right) $$

[13]. This operator identification is classical algebra. The Gamma basis selects and normalizes the relevant analytic null germs at $u = 1$ and connects them to the harmonic-tail coefficient matrix; that connection is the substantive result here.

8.9 Verification and the next finite obstruction

The accompanying scripts separate exact algebra from numerical diagnostics. Direct Gamma quadrature at 45 decimal digits for $m = 1, 2, 3$ and $1 \le n \le m + 3$ gives all 18 coefficient recurrence residuals below 10^{-46} and all six top-derivative identity residuals below 10^{-42}, with the normalizations recorded separately. A separate calculation evaluates T_m by Gamma quadrature at 60 digits, then propagates the exact recurrence to compare the determinant with (8.36). The recorded ratios are

	$b = 20$	$b = 40$	$b = 80$	$b = 120$
$m = 1$	0.951663	0.975424	0.987607	0.991714
$m = 2$	0.796107	0.891056	0.943647	0.961999
$m = 3$	0.456866	0.687818	0.832149	0.885309

Their approach to one is a consistency check on the explicit normalization; the infinite theorem is proved above independently of these finite computations.

The most direct continuation is to make the remainder in (8.36) effective, obtain a certified cutoff for each fixed m, and use exact rational arithmetic below it. The gap-six case is a natural first target because its initial sign transition rules out a blanket positivity shortcut. Further questions include exact evaluations of K_m, a classification of finite sign changes, and uniform asymptotics when m grows with b. The Gamma moments also suggest parameter deformations of (8.17) and (8.13); any such extension must recheck the contour residues and the original coordinate bridge.

9 Research status, verification, and next questions

9.1 What has been established

The proof-bearing deliverable is the article itself. The programs check finite algebra and independent numerical instances; they are not substituted for its convergence, continuation, contour, or asymptotic arguments. The principal advances can be mapped precisely to the source questions.

Source question	Result supplied here	Remaining scope
Individual ordered mixed labels	Theorem 2.1 gives the exact two-particle identity on an open complex domain. Theorems 3.1 and 4.4 give its continuation and all positive-order values.	Higher rank is handled by finite localization, not a universal single-zeta assertion.

Source question	Result supplied here	Remaining scope
Noninteger cutoff generators	Theorem 5.1 and (5.9) give a finite polylogarithm and beta–Lerch transform, entire in both orders on the disk.	The endpoint requires its own convergence domain and collective cancellations.
Other annihilating stencils	Section 7 proves a finite block formula for critical integer stencils and a stronger reduction under one further annihilation condition.	This is a sufficient family, not a classification of arbitrary noninteger grids.
All unequal even harmonic-tail prefixes	Section 8 proves eventual optimality on every fixed-gap ray, with exact Gamma normalization and a nonzero signed asymptotic.	No effective threshold is provided. The full all-pairs conjecture, including complete gap-six coverage, remains open here.
Stieltjes derivatives and primitives	Theorems 6.1 and 6.3 provide ordinary all-order identities. Equations (4.17) and (6.19) give compatible primitives.	Elementary symmetric arcsine cases are recurrence consequences; they are not independent priority claims.
Short Gaussian S_6 and revised S_8	Their present source status has been checked and preserved.	Neither identity is proved or refuted by this article.

The incoming bilinear and tail-kernel reports already supply complete smaller-order-ten and smaller-order-twelve reconstruction theorems [6, 7]. They should retain their existing attribution and audit status. Reproving or relabeling an already supplied finite case would not constitute a new research advance; the fixed-gap theorem instead changes which infinite direction is controlled.

9.2 Audit findings and a proposed correction

No newly false inherited mathematical theorem was identified in the material used for this article. There is a concrete editorial defect: the manuscript README repeats consecutively the complete paragraph describing the 1,032-input first-letter-X family, its rank 1,008, and its kernel dimension 24 [1]. The package includes a narrowly scoped patch removing the second copy. This is a proposed README correction, not a modification of a mathematical statement.

Several distinctions matter when the new results are integrated. The source already records that the old S_8 candidate was rejected and the revised candidate is still conjectural. The small residual enclosures for S_6 and revised S_8 establish proximity, not zero. Likewise, the canonical modular relation searches do not assert rational nonmembership or a counterexample to a period equality [2, 3]. These are inherited status safeguards, not newly discovered errors.

For the present contribution, the analogous precise statements are: the zero-weight limits in Section 3 concern a meromorphic continuation; the constant in (6.7) is not the fixed-coordinate Laurent constant; block formulas outside their separated convergence domain are interpreted collectively; and full rank of a formal moment matrix does not prove arithmetic independence of the evaluated constants. The Gamma argument also shows why positivity of the scalar

recurrence does not imply positivity of every minor: even modes lose their leading Laplace phase, and the determinant has the parity-dependent sign derived in Section 8.

9.3 Reproducibility and the independent checks

The package contains modular TeX, a self-contained assembled TeX file, the compiled PDF, a build command, a claim-status file, exact source locations, and portable verification programs. Source repository material was inspected at the pinned revision and kept unchanged. The generated correction patch is separate.

The ordered-boundary suite checks the finite summation-by-parts identity on 240 arbitrary rational sequence/cutoff cases, the rational partial-fraction identity on 36 order pairs, 570 parity coefficients, 420 coefficients of the cutoff generator, and all four entries of the zero-weight directional table. Four explicit infinite-sum values are also compared with directly accumulated original ordered sums, using both positive tail envelopes and separately labelled accelerated diagnostics.

The corollary suite checks 13 exact master-function derivatives and special values and nine exact formal Stieltjes coefficients. It compares the geometric master with the original resolvent-weighted ordered sum, tests two subtracted harmonic sums, follows six unaccelerated hard-cutoff limits, and evaluates nine arcsine integrals. The beta–Stieltjes suite checks five Bell–Leibniz coefficients, independent Stieltjes-value samples, and twelve numerical beta integrals through jet order two, including independently varied shifts. Its expansion and endpoint truncations are recorded separately from floating-point roundoff.

The localization suite supplies 147 exact checks, including finite-support transport, the full contact table, both rank-three two-index sectors, singleton cancellation through rank five, and a deliberately corrupted control. The Gamma suite compares directly integrated basis coefficients with the rational recurrence for $m = 1, 2, 3$, checks the exact inhomogeneous moment, and records finite-index ratios to the determinant asymptotic. A separate exact suite checks the Bernoulli matrix against its complementary-minor normalization and the harmonic correction. The independent proof audit rederives the contour shifts, both residues, the initial determinant, the parity-sensitive asymptotics, and the source bridge. This is internal independent checking, not external peer review or proof-assistant verification.

All numerical comparisons are diagnostics at their stated working precisions. The proof of eventual nonvanishing is the nonzero limiting Vandermonde determinant, not an extrapolation of the displayed finite data. In particular, a finite gap-six sign table does not provide the missing effective cutoff.

9.4 Further research questions

The following questions are arranged by how directly the present proofs supply a route forward. They concern exact identities and structural classification; estimates enter only where needed to turn an eventual identity theorem into a finite decision.

1. **Complete the fixed-gap prefix conjectures.** For each fixed m, turn the nonzero asymptotic into an explicit computable threshold $B(m)$ and combine it with exact rational determinants below that threshold. The specific next case is gap six. The exact values at $b = 4, 5, 6$ are negative, and the checked values $7 \le b \le 40$ are positive. The strengthened conjecture

$$\det F_{b,3} > 0 \qquad (b \ge 7)$$

is consistent with both the finite data and the proved eventual sign, but is not proved here. A successful completion must justify the transition from a finite table to all indices.

2. **Evaluate the Gamma normalization constants.** The positive integral K_m is exact, and $K_1 = 8\pi$. Find closed Gamma, hypergeometric, or modular descriptions of $K_2, K_3, \ldots$. The multiple-orthogonality interpretation suggests studying the associated moment determinant rather than integrating each entry separately. A recurrence in m, if it exists, would be particularly useful.

3. **Connect the two eventual regimes.** The new theorem fixes the gap while the earlier theorem fixes the smaller order. Can one obtain a two-parameter determinant expansion that covers the region between those rays? The limiting nodes $(j\pi)^{-2}$ become less separated as m grows, so a proof must control the full determinant, not merely reuse fixed-m error constants. This is a possible route toward the all-pairs conjecture.

4. **Classify the finite sign changes.** The exact Gamma representation is valid at every $b > m$, while its sign is presently asymptotic. Determine the number and locations of the finite sign transitions for each m. A variation-diminishing statement adapted to the polynomials $\mathrm{Im}(x + i/2)^{2n}$ could replace a false appeal to unqualified total positivity.

5. **Deform the Gamma basis.** Replace the quarter shift by a parameter and seek the corresponding shift equation, residue forcing, and reconstruction matrix. Identify which deformations preserve a finite set of sinh/tanh modes and which create additional residue terms. Differentiating a valid deformation would give new exact polygamma-weighted orthogonality identities; the deformation itself must be proved before its derivatives are used.

6. **Determine the actual higher-rank period coordinates.** The block theorem gives at most $\lfloor (r+1)/2 \rfloor$ free ordered indices. At rank three the two-index contact sector is nonzero, but this does not prove arithmetical depth two. Determine which integer orders, shift collisions, or label symmetries reduce these sectors to single Hurwitz values, and which require genuine nested periods. The explicit contact polynomial provides a finite starting point.

7. **Compute all-rank polar geometry from the blocks.** The two-particle oblique residues are explicit polynomials. At higher rank, the finite sector formula should organize polar hyperplanes by free blocks and their mergers. Prove a complete local Laurent description, including its behavior when the original finite-difference weights vanish. This would extend the directional table without confusing a pointwise specialization with a limit.

8. **Classify stencils by ordered boundary support.** The annihilation conditions proved here are sufficient. Seek necessary conditions for a prescribed maximum number of free blocks, first for finite integer stencils. A discrete-residue criterion might recognize additional cancellations invisible to degree counting. Nonuniform or noninteger grids require a separate analysis because there is no common integer transport lattice.

9. **Find new exact beta–Stieltjes evaluations.** For rational σ, special shifts may combine the beta weight with multiplication or reflection formulas. Determine when (6.11) reduces to a finite expression in zeta derivatives, log-Gamma values, and lower Stieltjes constants, rather than the generally infinite but geometric expansion (6.13). Confluent shift derivatives offer a systematic way to introduce polygamma factors.

10. **Construct higher-rank cutoff formulas.** Insert a common outer cutoff before transporting the stencils. Track the factors of $1 - z$ attached to free collision blocks and determine whether the endpoint depth reduction can be read directly from them. Section 5 proves this mechanism at rank two. A higher-rank formula should retain all finite endpoint terms before any limiting operation.

11. **Return to Gaussian periods with a new functional route.** The short S_6 and revised S_8 questions remain worthwhile, but the inspected finite relation spaces have already been analyzed carefully. New work should supply a functional identity, a genuinely larger motivic or iterated-integral relation, or a rigorously changed candidate. Further decimal agreement alone cannot settle either equality. The ordered kernels here may suggest new generating relations, but no such implication is presently proved.

9.5 Suggested integration

The ordered identity and its cutoff generator naturally continue the finite-difference report. The beta–Stieltjes identities belong with the Hurwitz and Stieltjes calculus, with their ordinary convergence hypotheses retained. The all-rank transport theorem can be kept as a separate structural continuation. The Gamma basis should follow the exact fixed-gap determinant reduction in the harmonic-tail Gamma branch, with its inherited coordinate convention printed explicitly.

The accompanying integration note lists these destinations, theorem dependencies, a proposed status update, and the README patch. Canonical acceptance should record the proved fixed-gap eventual statement at its actual scope; it should not silently promote the remaining full reconstruction or Gaussian conjectures.

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[2] ProveIt Contributors, *Editorial Ledger*, canonical claim and integration statuses, same repository snapshot. Pinned editorial ledger.

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[6] ProveIt Contributors, *Bilinear Tails and Harmonic Determinants*, incoming report, 11 October 2026; rational boundary certificates, cutoff jets, and order-ten reconstruction. Pinned incoming archive.

[7] ProveIt Contributors, *Tail Kernels and Boundary Products*, incoming report, 11 October 2026; Sections 6 and 7 supply complete smaller-order-ten and smaller-order-twelve reconstruction theorems. Pinned incoming archive. All seven incoming archives and their specific roles are listed in the accompanying source map.

[8] ProveIt Contributors, *Quartic Spectral Reconstruction*, incoming report, 11 October 2026; Section 3 proves eventual reconstruction with the smaller order fixed, and Section 8 poses the order-ten/twelve tasks subsequently addressed in the other incoming reports. Pinned incoming archive.

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